



MAIZE VALUE CHAIN IN KENYA: IMPLICATIONS OF MAIZE MILLING COSTS

Report

January, 2020



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Abbreviations and Acronyms

AFA	Agriculture and Food Authority
CGA	Cereal Growers Association
COMESA	Common Market for Eastern and Southern Africa
EAC	East African Community
EAGC	Eastern Africa Grain Council
FAO	Food and Agriculture Organisation of the United Nations
FAOSTAT	Statistical Database of the Food and Agriculture Organisation of the United Nations
FEWSNET	Famine Early Warning System Network
IGC	International Grain Council
IPP	Import Parity Price
IPPC	International Plant Protection Convention
ITC	International Trade Centre
KAM	Kenya Association of Manufacturers
KES	Kenya Shilling
KNBS	Kenya National Bureau of Statistics
KOMEX	Kenya Multi-Commodity Exchange
NCPB	National Cereals and Produce Board
NEXT	Derivatives Market of the Nairobi Securities Exchange
RATIN	Regional Agricultural Trade Intelligence Network
SFRTF	Strategic Food Reserve Trust Fund
UN COMTRADE	United Nations COMTRADE Database
USD	United States Dollar
USDA	United States Department for Agriculture
VAT	Value Added Tax
WRS	Warehouse Receipts System





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Foreword

Maize is the staple food for most Kenya households, with a per capita consumption of 60 kgs, which is equivalent to 5kg per person, per month, as of 2018.

Despite this, maize production faces several challenges, including a widening yield gap due to low input use and post-harvest losses along the value chain. This hampers the realization of our aspirations to enhance food security and nutrition in the country.

The government has earmarked maize production as a priority value to ensure broad-based socio-economic growth in the agriculture sector. Taking a full value chain approach, from farm to table or farm to fashion is critical to realizing this transformation. Through this approach, we are in a position to bring in expertise, technological and financing capacities to increase agricultural production. In turn, there will be a constant supply of products for value addition.

Value addition facilitates forward and backward linkages with other economic sectors, therefore, creates employment and wealth along the value chain. Such linkages are especially stronger in manufacturing than in other sectors. Raising maize productivity and linking it to manufacturing through agro-processing and agro-based manufacturing will not only drive growth in the agricultural and industrial sectors but also create equitable and sustainable growth for our people.

The government has been keen on transforming the agricultural sector as outlined in the *Agricultural Sector Transformation and Growth Strategy*. The strategy seeks to tap into agriculture value addition to serve both domestic and export markets and utilizing Kenya's intrinsic production potential.

This is also highlighted in the Big 4 Agenda, which presents a huge opportunity for agricultural transformation particularly through two pillars; Food Security and Manufacturing. The Food Security pillar seeks to expand food production and supply, and

reduction of food prices to ensure affordability. It also looks at supporting value addition in the food processing value chain. On the other hand, the Manufacturing pillar seeks to add value to food processes, among others processes, and to raise the manufacturing sector's share to GDP to accelerate economic growth, create jobs and reduce poverty.

Proper understanding of the challenges hampering maize production is fundamental to the creation of policies and strategies that not only reduce costs for players within the value chain but also provide an environment that promotes investments, technological advancements and competitiveness for sector players. For instance, what role can public-private partnership play in reducing post-harvest losses and improving grain handling in the value chain? What policy reforms can reduce the cost of milling in the country? What regulatory and compliance improvements do we need for food safety in the maize value chain?

This report seeks to give answers to these questions as it sheds light on how we can enhance maize production whilst lowering milling costs. It also seeks to highlight the role of maize, as one of the cheapest sources of calories for many households, in contributing to food security, trade and livelihoods of millions of people in Kenya. We remain hopeful that this report will contribute to a flourishing maize value chain in the country.



Mucai Kunyihya
KAM Chairman

Mucai Kunyihya
Chairman, Kenya Association of Manufacturers



Executive Summary

Demand and Supply

- Maize plays a central role in contributing to food security, trade and livelihoods of millions of people in Kenya. Maize is the cheapest source of calories among the Kenyan households accounting for 24 per cent of all calories consumed and 50 per cent of staple food calories consumed.
- The demand for maize currently stands at 52 million bags, and it has been projected to reach 60 million bags by 2025. Kenya is a net importer of maize due to higher consumption relative to domestic production. The average maize production over the past 15 years has been estimated at 3.2 million metric tons, while consumption was 3.5 million metric tons.



Phyllis Wakiaga
KAM Chief Executive

Gross Margin analysis for maize production

- Large-scale farmers are 33% more productive than small-holder farmers, with large-scale farmers achieving yields of 24 bags per acre compared to 18 bags per acre by smallholder farmers¹. This means that large scale farmers earn almost 39% higher profits than their smallholder counterparts despite incurring 31% higher production costs.
- The three biggest cost drivers for maize production irrespective of the production system are fertilisers, machinery and labour. Collectively, these costs account for 64% of total production costs in large-scale systems and 77% in smallholder production systems. However, labour is a far bigger cost driver for smallholder farmers (36% of total production costs) compared to large-scale farmers, most likely due to limited mechanisation in smallholder farming systems.

Maize trade

- Cross-border trade flows are aligned to the respective crop calendars for Kenya and its main trading partners (Uganda and Tanzania). Peak imports are usually during the second and third quarters of the year, which coincide with the lean period domestically and harvests from trading partners. Imports from Uganda typically peak in the third quarter while those from Tanzania peak during the second quarter.
- Informal cross-border trade usually surpasses the volumes that are sold through the formal channel. For instance, in 2017, Kenya informally imported 368,167MT of maize from Uganda. Kenya, like the other EAC states, has adopted the harmonized EAC standards for maize, however, implementation and enforcement is limited.

¹ Tegemeo (2018) define a large-scale farmer as one who farms 50 or more acres of maize, while a smallholder farmer cultivates less than 10 acres of maize. Both types of farmers have 1 season of maize



Post-harvest Losses

- Post-harvest losses in the maize are estimated at 8% - 40% according to different reports. Regarding the value chain structure as a percentage of total production, most losses happen at storage (9%) due to pest infestation, moulding caused by inadequate drying, rodents and poor hygiene. Smaller amounts are also lost during shelling (1.8%), drying (3%) and harvesting (2.2%).
- Maize flour has been implicated in aflatoxin poisoning resulting from mould contamination of maize on the farm or in storage. Hermetic storage bags, designed to reduce insect pest losses, have been shown to reduce aflatoxin contamination by between 40% and 66% after three months of storage, depending on the starting moisture content, level of contamination and other environmental factors.

Maize Milling

- Maize milling is dominated by small and medium scale millers, who are estimated to control approximately 60% of the market, with an approximated milling capacity of not more than 150MT per day. Large-scale millers account for approximately 40% of the market with an installed milling capacity of approximately 6,600MT per day.
- Hammer mills (posho millers) have a milling capacity of 6 - 110 metric ton per month, depending on the location of the mill. In addition, some posho millers had invested in maize dehullers to produce refined flour commonly referred to as "Number 1". These mills are mainly found in rural areas and urban slum areas. The main challenge facing these millers is limited capital for expansion. Additionally, most hammer millers use diesel as the source of energy which increases the cost of operation, considering that price of diesel is regulated and varies every fortnight. Due to the volumes that are milled, there is underutilization of the fixed costs, such as machines and labour.
- Small-scale millers are estimated to be about 50–75 in number with an installed capacity ranging from 10 - 200 metric tons per month. These millers produce sifted maize meals and maize by-products such as the composite maize meal that they provide in their own consumer brand packaging. The major constraints faced by the small-scale millers is the use of poor technologies which affects the quality of their products. The other constraints are underutilized processing capacities, limited access to finance, unreliability of grain supply and high cost of power.
- Medium and large-scale millers account for about 93% of the annual total installed milling capacity, estimated at 1.4 million metric tons. Most of the mills are located in Nairobi and Mombasa. Both cities account for 60% of the national installed capacity, followed by Eldoret with 15%, then Thika, Kisumu and Kitale with the remaining installed capacity. The four main large-scale millers account for 70% of total milling capacity. Apart from sifted maize meal, these millers are also involved in animal feed production.
- Cost of milling maize grain: The cost of maize grain accounts for approximately 75% of milling costs, packaging costs (about 8% of milling costs) and handling costs which include cleaning, drying, storage and quality control (7% of milling costs).
- High cost of post-harvest handling and storage: For instance, for farmers in Nakuru, mechanical drying accounts for 56% the post-harvest handling costs as opposed to sun-drying, which accounts for 34%. Sun-drying is only practical for small quantities of grain. This means that a farmer will incur a cost of over KES 500 per bag on storage and handling if they harvest maize with a moisture content of 20%.
- Low enforcement of quality standards for maize grain. Maize grain in the domestic market is not priced based on quality consideration. In most cases, grain is affected by moisture content and aflatoxin contamination, especially during harvest seasons, that coincide with heavy rainfall. Other challenges limiting enforcement of standards include the length of time taken to receive test results from laboratories, the cost of the tests and the cost of compliance.
- Cost of doing business for millers has increased over the past five years, driven mostly by substantial increases in regulatory compliance costs (taxes, fees and permits), costs of compliance with standards (sourcing quality grain and meeting standards requirements for milled products), government participation in markets through food reserves and increase in costs of cross-border trade despite the reduction in time taken to clear goods across borders.



Recommendations

The study proposes the following to reduce the cost of milling maize grain:

- i. Improve the market system for maize trade through structured trading systems. Such as the Kenya Multi commodity Exchange (KOMEX), the NeXT Commodity Derivatives Exchange of the Nairobi Securities Exchange and the EAGC G-SOKO Grain Trading Systems. Adoption of these structured trading systems for grain trading will improve efficiency in maize grain trade. The systems should be supported by the Warehouse Receipt System (WRS) that offers millers cost reduction, quality and quantity guarantee, reduced multiple handling of grains and price risk management and procurement planning.
- ii. Reduction of taxes imposed on maize grain. Specially, allowing duty-free yellow maize specifically for the animal feed under duty remission scheme. Allow tax-free transaction for purchases of post-harvest handling equipment, materials and services. Abolishing multiple cesses and domestic tariffs incurred during transportation and handling of maize consignments. Concerted advocacy efforts by grain industry stakeholders are thus required, particularly through the annual national budget formulation processes, to remove these taxes.
- iii. Promote Public-Private Partnerships (PPPs) to reduce post-harvest losses and improve grain handling and marketing in the maize value chain through restructuring NCPB into a public-private commercial entity. Additionally, engaging the private sector for storage and handling of food stocks on behalf of the strategic food reserve.
- iv. Strengthen data collection and information sharing to inform policy and business decisions.
- v. Strengthen implementation and compliance mechanisms of quality standard for improved food safety along the maize value chain.

Phyllis Wakiaga

Chief Executive, Kenya Association of Manufacturers



1. Introduction





1 Introduction

1.1. Background

Maize plays a central role in household food security and trade. It is a source of livelihood for millions of Kenyans. Maize is annually grown on an estimated 2.2 million hectares by large-scale farmers (25%) and smallholders (75%) as both commercial and subsistence crop (Kirimi et al., 2011). On average, approximately 40 million bags of maize are produced annually which accounts for 84% of total cereals production in Kenya (KNBS, 2018). The national demand for maize is estimated at 52 million bags annually whereas per capita consumption is estimated at about 67.3kg per year (KNBS, 2020). Therefore, the annual deficit of 12 million bags is met through trade with neighbouring countries such as Uganda and Tanzania. Maize imports have increased by more than 15% between 2014 and 2018 (KNBS, 2019). The country's self-sufficiency level for maize was the highest in 2013 where it reached 97% and lowest in 2017 (70%) due to the drought and floods episodes (KNBS, 2018).

Maize production in the country is constrained by several challenges, namely overreliance on rainfed production thus vulnerable to weather variability, a widening yield gap due to low input use, coupled with pest and diseases incidences (maize lethal necrotic diseases and fall armyworms). Additionally, post-harvest losses continue to be high ranging from 8% - 40% along the value chain, inadequate storage facilities, unpredictable producer prices and unstructured markets (GoK, 2019). In the agriculture sector, maize has been earmarked as a priority value to ensure broad-based socio-economic growth. The government strategic agenda over the next five years (2018-2022) under the food and nutrition security pillar has targeted to produce over 67 million bags of maize by 2022, this is more than a two-thirds increase from current production levels. The Agricultural Sector Transformation and Growth Strategy (ASTGS 2019-2029) identifies maize as one of the strategic agricultural commodities value chains and has made provisions to improve the production system through revamping input use and the marketing structure (MoALF, 2019).

While many studies have been carried out on the maize value chain, few have assessed the value chain from a business perspective to understand how cost affects the state of play for value chain actors. Without understanding the business perspective, which ultimately informs the day-to-day decisions of value chain actors, it becomes difficult to devise practical and sustainable solutions to challenges facing the maize sub-sector. Therefore, this study seeks to get an understanding of the various dynamics in the maize value chain and their implications on value chain players, with a particular focus on millers.



1.2. Objectives of the Study

The main objective of the study is to examine the maize value chain in Kenya to identify opportunities and challenges. Specific objectives were to:

- i. Map out the maize value chains and identify the types of organisations in primary production, secondary production and storage facilities.
- ii. Provide background information on the significance of the maize value chain in Kenya to imports, exports and subsequent value addition in the country.
- iii. Identify and expound on the bottlenecks in the maize value chain including soil conditions, use of right seeds and fertilisers, better crop husbandry, aflatoxin and the lack of extension services to farmers.
- iv. Document post-harvest losses occurring in the maize value chain, corresponding approximate economic losses and propose possible interventions.
- v. Conduct a case study on the National Cereals and Produce Board (NCPB), mapping available silos, locations and their capacities and propose a framework for public-private improve grain handling and marketing in the maize value chain.
- vi. Document the position of manufacturers/millers and the impact of the post-harvest losses on product costing.

1.3. Research Methodology

1.3.1. Value Chain Approach

Value chain analysis has gained considerable importance in recent years. Mainly because of the need to assess the key sources of cost efficiency or the lack of it along the value-chain of the commodity to come up with appropriate policy interventions aimed at raising overall value-chain efficiency. The United Nations Industrial Development Organisation (UNIDO)(2009) describes a value chain as the entire range of activities that take place to bring a product from the initial input-supply stage, through the various phases of processing, to its final market destination, including its disposal after use. For instance, agro-food value chains encompass activities that take place at the farm level, including input supply, and continue through handling, processing, storage, packaging and distribution. As products move successively through the various stages, transactions take place between multiple chain stakeholders - money changes hands, information is shared - thus, value is progressively added. Hence a value chain is a system of interdependent activities.

1.3.2. Sampling

A purposively random sampling approach was used to select the value chain players (farmers, traders, processors). The sample was drawn from a list of EAGC's Members and stakeholders.

Primary Data Collection

Data was collected through correspondence with maize value chain players, using a semi-structured questionnaire whereby the respondents filled out the survey and sent it back to the study team. Annex 1 summarises the respondent groups and the type of data and information collected from each of them.



Secondary Data Collection

Existing research papers, reports and policy documents were reviewed including:

Table 1: Documents and papers reviewed during the study

Class of documents	Documents reviewed
National Policy documents	• Agricultural Sector Transformation and Growth Strategy • The Big Four Agenda • Vision 2030
National Laws and Regulations	• Warehouse Receipt System Act • The AFA Act • The Crops Act • The NCPB Act • The Public Finance Management (Strategic Food Reserve Trust Fund) Regulations, 2015 • The Standards Act
Regional Policy Documents	• The EAC Agriculture and Rural Development Strategy • The EAC Food Security Action Plan II • The EAC Food and Nutrition Security Strategy • The EAC Aflatoxin Prevention and Control Strategy • The EAC CAADP Regional Agriculture Investment Plan
Maize sector studies in Kenya	• Annual Cost of Production surveys by Tegemeo Institute • Maize Taskforce report 2018 • Report of Proceedings of the Food Security Experts Forum 2018
Business Environment studies/reports	• World Bank Enabling the Business of Agriculture Report 2019

Source: Authors compilation

1.3.3. Data Analysis

Data was collected and analysed quantitatively. Qualitative analytical methods included the compilation and classification of the collected information using a Likert scale.



2. Global and Regional Overview of the Maize Sector



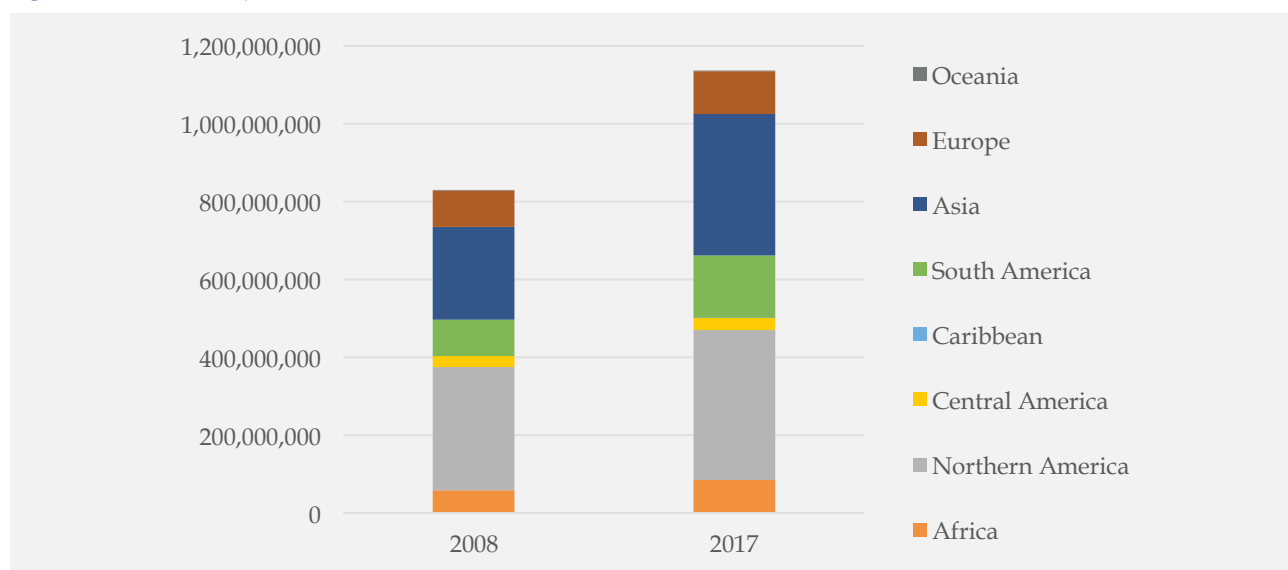


2 Global and Regional Overview of the Maize Sector

2.1. Production Trends

Maize is the leading global cereal in terms of production, with 1.1 billion metric tonnes (MMT) produced globally (FAOSTAT, 2017). Maize is produced across temperate and tropical zones spanning all continents. Its production increased by 37% from 2008 to 2017 (Figure 1). (FAOSTAT, n.d.).

Figure 1: Global maize production, 2008-2017

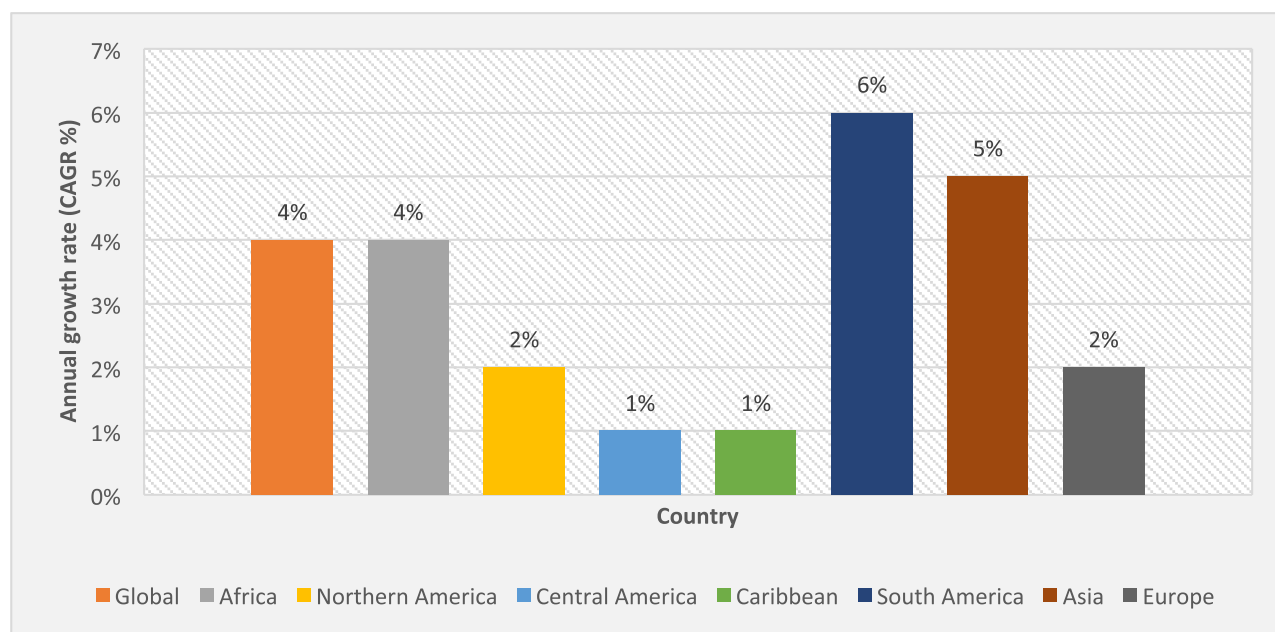


Source: FAOSTAT, 2019

The increasing demand for maize and its global advance implies that by 2023, maize will account for the highest share (34%) of the total crop area harvested (OECD-FAO, 2014). Biotech maize occupied 58.9 million hectares globally, which was 31% of the global maize production in 2018 (ISAAA, 2018). Implying there will be challenges maintaining the required volumes of non-genetically modified maize in the world market available to particularly low- and middle-income countries where the growth in production for maize averaged at 4% between 2008 and 2017 (Figure 2). The International Grain Council (IGC) expects global output to grow by 9% over the next five years, driven by productivity gains and increased acreage particularly in the U.S. where farmers are shifting from soybeans to maize. Production in Africa is expected to be driven mainly by increased acreage (IGC, 2019).



Figure 2: Maize production growth by region, 2008-2017

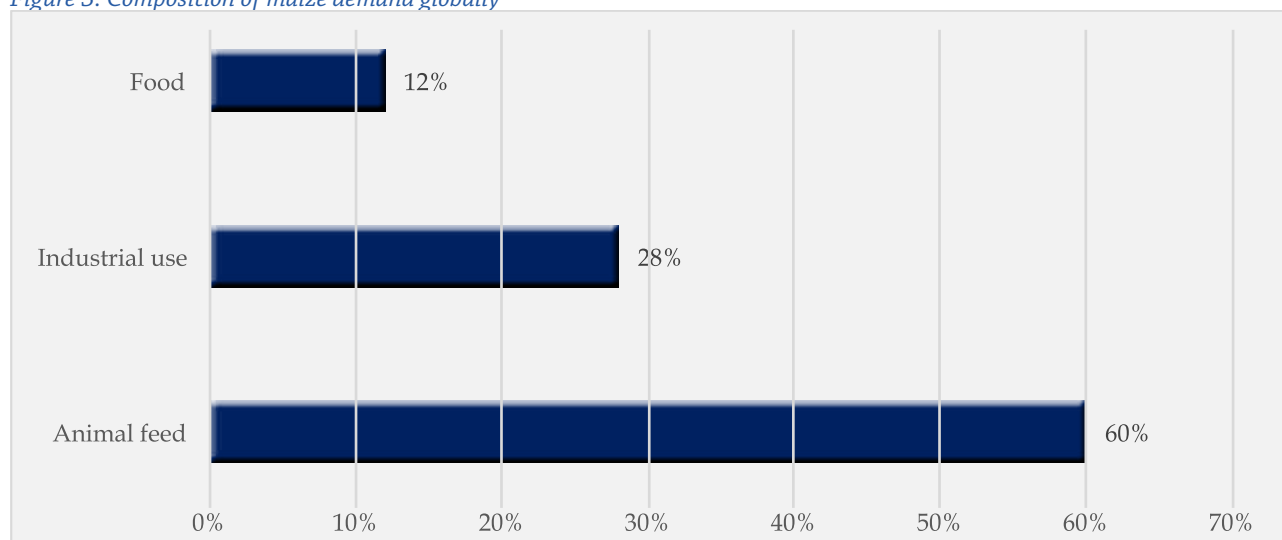


Source: FAOSTAT, 2019

2.2. Consumption Trends

The predominant use of maize globally is as an ingredient in animal feed, which accounts for approximately 60% of maize demand globally². Food accounts for only 12% of global maize (mainly in Africa), while industrial use (production of ethanol) accounts for 28% of the demand (Figure 3). The study estimates that cumulative global maize production is worth approximately USD 192 billion³. The demand for maize for animal feed, industrial use and food consumption is worth about USD 115.5 billion, 53.9 billion and 23.1 billion, respectively⁴.

Figure 3: Composition of maize demand globally



Source: Computed from IGC data, 2019

² Computed from IGC data.

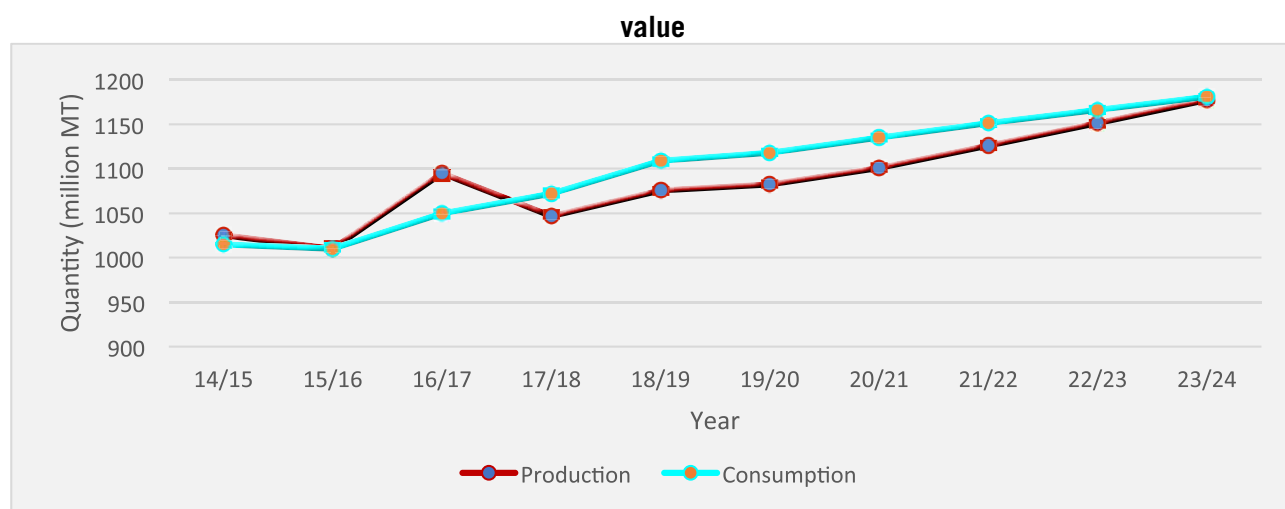
³ Global production assumed to be 1.1 billion metric tonnes annually as per IGC and FAO estimates. An average price of \$175/MT is used based on the highest and lowest prices recorded by IGC from December 2018 to December 2019.

⁴ Values based on 60% use in feed, 28% in industrial use and 12% for food as per IGC data and assumption in (2) above.



Population growth, changing diets and a rapidly growing poultry sector are contributing to a sharp increase in maize demand. Consumption growth, particularly outside Africa, is likely to be outpaced by production growth over the next five years, leading to a convergence of the two trends by 2023/24 (Figure 4). This is generally, attributed to slower output growth in industrial use, particularly ethanol production.

Figure 4: Global demand and supply trends and projections



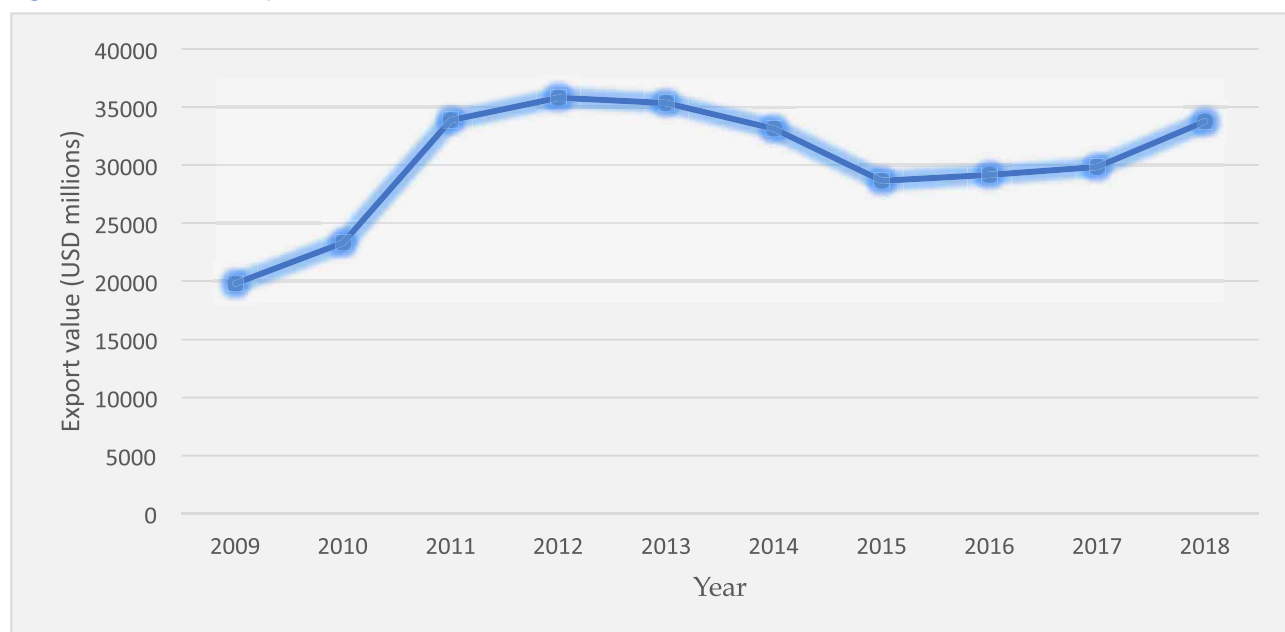
Source: IGC, 2019

2.3. Maize Markets and Trade

Global Maize Trade

Global maize trade has been growing steadily over the past decade. The value of exports has increased by 70% between 2009 and 2018, reaching USD 33.7 billion in 2018 (Figure 5). The key drivers for growth in trade are increased demand driven by population growth in Asia and Africa, which in turn fuels demand food and animal feed.

Figure 5: Global maize exports, 2009-2018



Source: UN COMTRADE, 2019



The USA, Brazil, Argentina and Ukraine are the four biggest maize exporters with a combined market share of 73% of global maize exports in 2018. On the other hand, Mexico, Japan, South Korea, Spain, Iran and Egypt are the biggest importers, together accounting for 40% of global maize imports in the same year. Egypt is Africa's biggest importer of maize (8.6 million MT in 2018), while South Africa is the biggest exporter on the continent (2.2 million MT in 2018).

Regional Maize Trade

In Eastern Africa, Ethiopia, Uganda and Tanzania are net exporters of maize while Kenya, Burundi, Rwanda and South Sudan are net importers. Furthermore, the regional crop (harvest) calendar, as illustrated by Table 2 below, exerts significant influence on trade patterns, particularly when harvests in surplus maize producing countries coincides with lean periods in deficit countries, thus creating an opportunity for trade.

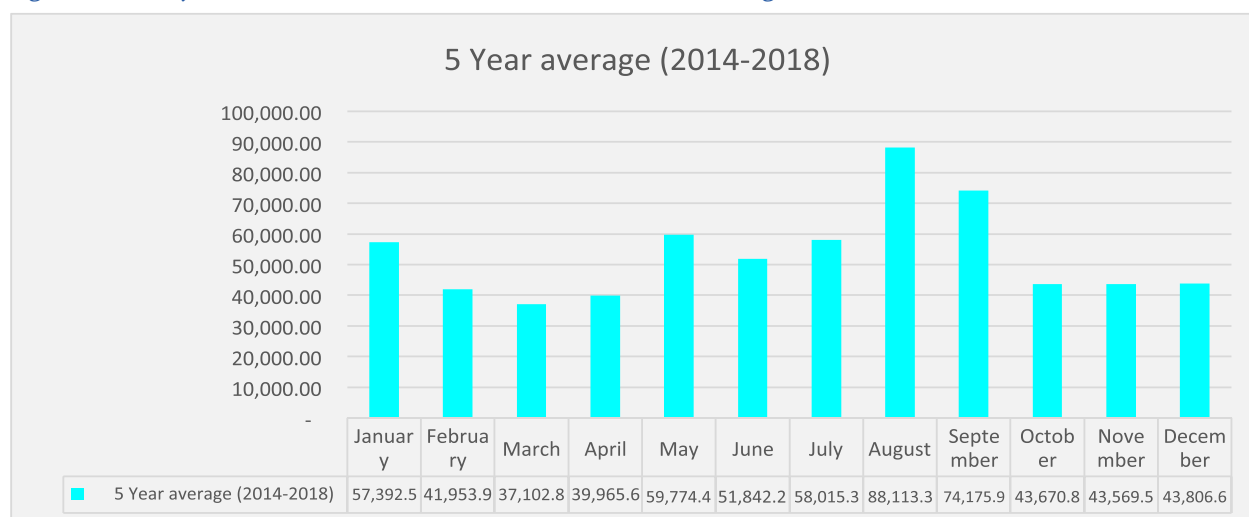
Table 2: Regional calendar for main maize harvest season

	J	F	M	A	M	J	J	A	S	O	N	D
Kenya												
Tanzania												
Uganda												
Burundi												
Rwanda												
S/Sudan												
Ethiopia												

Source: EAGC, 2019

Figure 6 illustrates monthly Eastern Africa cross-border trade volumes on the 5-year average basis. It demonstrates that trade volumes are highest during the second and third quarters of the year when the surplus countries harvest maize, which coincides with the lean period in Kenya, while trade flows are lowest in the first and fourth quarters due to suppressed demand from deficit countries.

Figure 6: Monthly sum of formal and informal cross border trade of maize grain in main trade corridors in Eastern Africa



Source: EAGC and FEWSNET, 2019

A typical value chain for maize grain can be summarised into four key nodes, namely production, aggregation and trade, processing and consumption (Figure 7). Farmers (small and large) are the primary producers. The farmer produces the maize through a combination of natural resources (land and water), machinery and labour, and agro-inputs (seeds, fertilisers and pesticides). Upon harvesting, farmers sell their produce – either individually or collectively – to traders. Traders provide a critical link between farmers and processors.

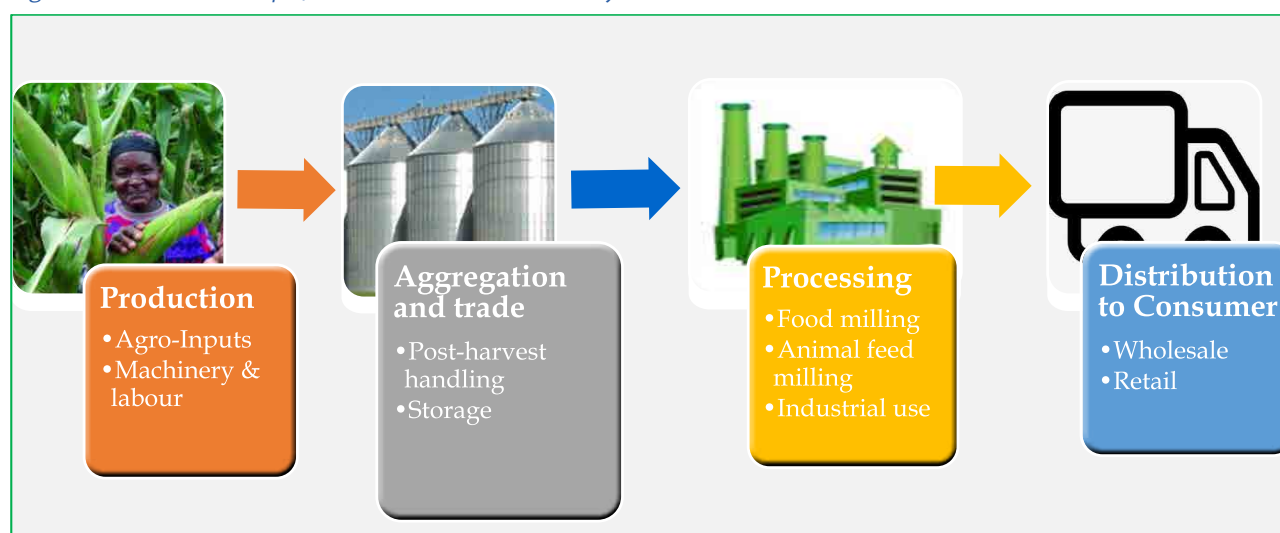


At the trade level, the commodities are usually aggregated, conditioned (cleaned and dried), and either offered for immediate sale to millers or stored in anticipation of more favourable market movements. Traders, large- and small-scale farmers sell their commodities to the Strategic Food Reserve Trust Fund (SFRTF) through the National Cereals and Produce Board (NCPB). Processors - secondary producers of food and animal feeds - usually procure their stocks from private traders for milling purposes. During shortages, millers also purchase stock from the SFRTF through NCPB, often at discounted prices. In Kenya, the use of maize in the manufacture of industrial products such as ethanol is insignificant.

Additionally, key things to note about the maize value chain include:

- Smallholder farmers dominate production. However, smallholder farmers have wider yield gaps compared to large scale farmers; the latter enjoy economies of scale, which allow cost reduction per unit and higher yields.
- There are two primary trade levels. The first level is small traders who buy directly from farmers. These traders pay cash for maize purchases and are the primary source of liquidity for farmers who have an urgent need for money. The second trade level is the larger traders who not only buy from small traders locally but may also source grains from neighbouring countries or international markets. Some may also act as buying agents on behalf of millers.
- Millers typically decide the price based on daily supply and demand. If the quantities are high, millers may suppress the price and vice versa.

Figure 7: Summarised map of the maize value chain in Kenya



Source: EAGC, 2019



3. Mapping the Maize Value Chain in Kenya



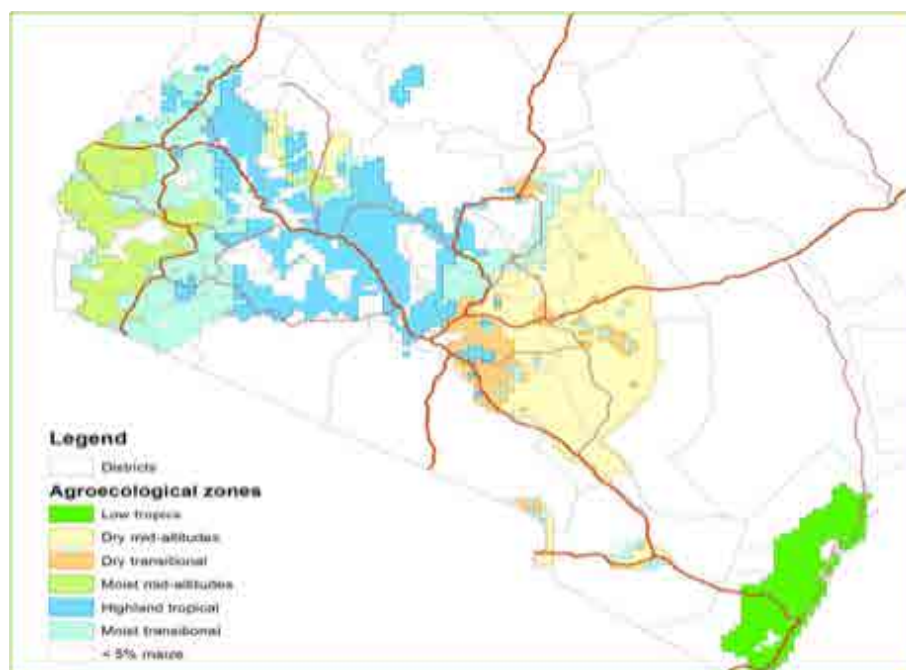


3 Mapping the Maize Value Chain in Kenya

3.1. Production

Maize is grown across all the major agro-ecological zones (AEZs) in the country. Counties in the Rift Valley region are by far the largest producers of maize in Kenya. They account for more than 51% of the total production. Maize is grown by small-, medium- and large-scale farmers, with small-scale farmers constituting the largest share of maize produced (Kiriti et al., 2011; ERA, 2013). Counties in the Rift Valley region are the largest producers of maize in the country, accounting for more than 51% of the total production, followed by counties in Nyanza and Western regions. Counties in the Central and Coast region contribute about 6% and 4% of the total output, respectively. Nairobi county and counties in the North-Eastern region account for less than 0.1% of the total production. As shown in Figure 8, maize production takes place in AEZs IV, V, and VI (Drought Tolerant Maize for Africa, 2015).

Figure 8: Maize agro-ecological zones

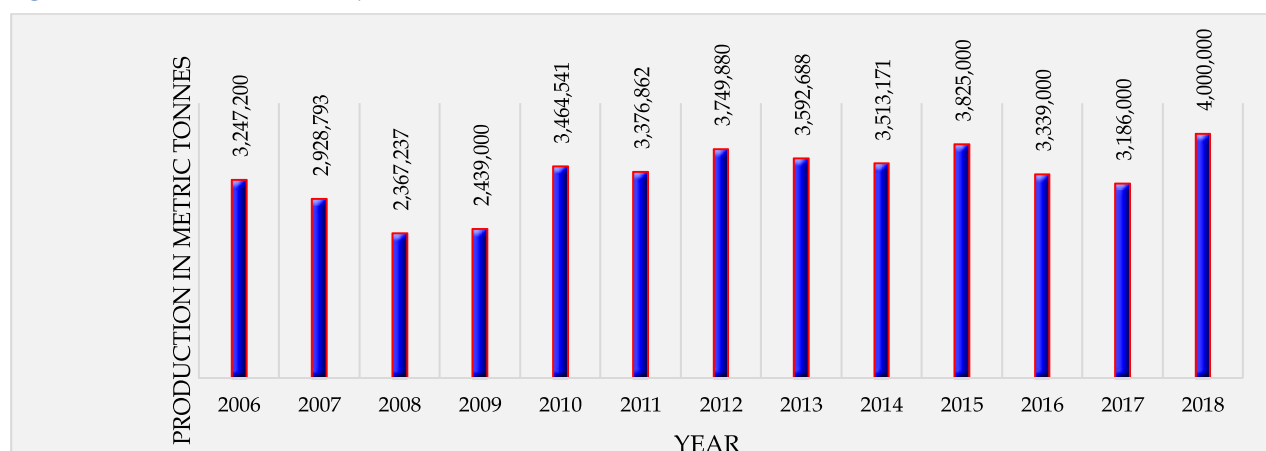


Source: Drought Tolerant Maize for Africa Bulletin, 2020



In East Africa Community (EAC), Kenya is the second leading producer of maize behind Tanzania and ahead of Uganda. In 2018, Kenya produced approximately 4 million metric tonnes of maize (Figure 9). However, Kenya is a maize deficit country and importation is required to account for the deficit. Imports come from Uganda and Tanzania to sustain domestic consumption during the low seasons.

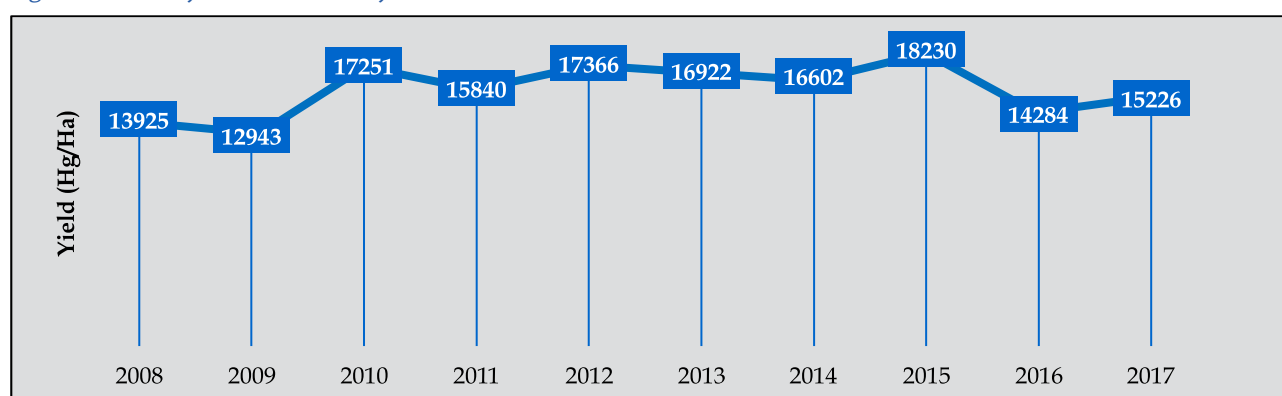
Figure 9: Maize Production in Kenya, 2006-2018



Source: FAOSTAT & USDA, 2020

Data from FAO (Figure 10) indicates that while yields have increased since 2008, they have plateaued and even declined in recent years. The subsequent stagnation and decline in yields are attributed to weather variability, outbreaks of pests and diseases - the Maize Lethal Necrosis disease in 2011 and Fall Armyworms in 2016/2017 - and declining soil fertility (MoALF, 2019).

Figure 10: Maize yield trends in Kenya, 2008 - 2017



Source: FAOSTAT, 2019

The major product at this node is mainly maize grain. The three categories of producers involved in production;

Small-Scale Farmers

The majority of the small-scale farmers in Kenya cultivate maize. These farmers cultivate less than 2 hectares of land and their production is characterized by small per capita volumes that result from the low use of inputs, low yields, limited land, and lack of storage facilities, among other factors. Small-scale farmers have limited access to critical services such as finance and warehousing. They sell their maize immediately after harvest to mostly small-scale traders or consuming households (Nyoro et al., 1999; Kirimi et al., 2011). These farmers can be categorized based on their purchase and sales characteristics. Over half of the small-scale farmers usually purchase maize and never sell it. Only 18% of the small-scale farmers purchase and sell their maize, and two-third of these farmers are net buyers of maize. In addition, the net buyers sell their maize at low prices during harvest and buy at a higher price during the lean season. Hence, this group is susceptible to volatile and high food prices, despite being producers. The warehouse receipt system has been mooted as a solution, where farmers store during harvest and sell when prices are high. However, the costs are prohibitive for the small-scale farmers (Coulter, 2009).



Medium-Scale Farmers

These farmers cultivate between 2 to 10 hectares of land and sell almost half of their production. The majority sell their maize produce to wholesalers, millers and the NCPB. Unlike the small-scale farmers, these farmers store their maize and sell it during the lean season. On average, they store their maize for 4–5 months before selling. In addition, these farmers have access to critical services such as financing and warehousing (Kirimu et al., 2011).

Large-Scale Farmers

Large-scale farmers cultivate over 10 hectares of land, purely for commercial purposes. Their farming is highly mechanized and input-intensive. They sell their produce to large-scale traders, the NCPB and millers. It is estimated that 60% of the national marketable maize in Kenya is sourced from these farmers and they store their maize for longer than 5 months to sell when prices are at their highest. These farmers benefit from the economies of scale; hence, they are able to access cheaper inputs, critical services (finance and warehousing) and negotiate better market prices. Large-scale farmers have formed a powerful political lobby group that has been the driver of maize policies implemented in Kenya in the past decade. Additionally, they have been the main advocates of subsidized fertiliser, from which they benefit the most (Kamau et al, 2013; Opiyo et al., 2015).

3.1.1. Input Supply

The most important inputs in the maize value chain are land, seeds, fertilisers, agrochemicals (Herbicides, fungicides and pesticides), farm equipment, water and irrigation equipment. Other services in the pre-production phase include extension services, market information, credit and certifications for production. Due to data challenges, only seed and fertiliser are highlighted briefly.

Most maize seed in the country comes from the formal sector, which has been one of the early success stories of breeding programs, with the launch of several widely adopted hybrids in the 1960s and 1970s leading to an 80% adoption of improved seed varieties (Das et al., 2019; Walker and Alwang, 2015). The national variety list constitutes 363 maize varieties, of which, over half were released in the last 10 years (KEPHIS, 2019). There has been a steady increase in the number of seed companies operating in the country in the last 10 years, however, the market continues to be dominated by Kenya Seed Company, a parastatal that holds a market share of 70–80% (Christinck et al., 2018). The volume of certified seed produced has increased from 27,078 MT in 2008 to 54,555 MT in 2017 according to KEPHIS. Despite this growth, the country still imported an estimated 7,000 MT of seed in 2017 from Zambia (KEPHIS, 2019).

Farmers in the country use an average of 30kg/ha fertiliser, which is far below the 50kg/ha recommended under the Abuja Declaration of 2006. Empirical evidence suggests that fertiliser use is rising rapidly, although this trend is concentrated in certain agro-ecological regions. There is increasing use of nitrogen-based fertiliser. Despite increased fertiliser use, the yields from smallholder farmers have been stagnant and well below what is obtained by many commercial farmers. Most farmers lack the financial resources to purchase enough mineral fertilisers to replace soil nutrients. The situation is further aggravated by the fact that even the farmers using the inorganic inputs hardly use the recommended rates (60 kg/ha) with most of them applying less than 20 kg/ha (Marenja and Barrett, 2009; Mutegi et al., 2012). Meru South District to determine the levels of complementarity between organic and mineral N amendments on maize yields and their influence on soil chemical properties. The experiment was set in a complete randomized block design (CRBD).

The country's fertiliser market is fully liberalized with the bulk of fertilisers imported and distributed by the private sector. In pursuit of the Kenya Vision 2030 and following the escalated world fertiliser prices in 2008, the Ministry of Agriculture established a fertiliser subsidy program to empower poor smallholder farmers engaged in maize production through the provision of farm inputs grants to support their normal livelihood activities. For 2018/2019 period, 635,000 bags were procured and another tender of 668,000 bags procured for the second-round totalling to 1.3 Million bags.

Some of the challenges associated with the fertiliser subsidy programme include:

- In most cases, the depots are placed far from the ordinary farmers reach
- Delayed procurement due to exchequer releases



- Delays in delivery of inputs to farmers
- Too much paperwork and inadequate capacity to track beneficiaries
- Farmers vetting does not take place

During the year 2020 cropping season, the Agriculture ministry suspended the subsidised fertiliser programme and rolled out the e-voucher fertiliser program, which will streamline the distribution and targeting of the beneficiaries of the subsidised fertiliser. According to a study carried out by Tegemeo Institute of Egerton University (2016), limited access to affordable credit and information coupled with poor infrastructure reduces the number of farmers who use fertiliser and certified seeds. Poorly developed inputs markets inhibit the use of improved technologies such as fertilisers, drought and disease-resistant varieties and mechanization, thus contributing to low productivity which is a problem faced by many farmers in Africa (AGRA, 2013).

3.1.2. Gross Margin Analysis for Maize Producers

Table 3 presents the gross margin analysis for maize production in Kenya. The analysis reveals that large-scale farmers are 33% more productive than small-holder farmers, with large-scale farmers achieving yields of 24 bags per acre compared to 18 bags per acre by smallholder farmers⁵. This means that large-scale farmers earn almost 39% higher profits than their smallholder counterparts despite incurring 31% higher production costs.

Table 3: Gross-margin analysis of maize production in Kenya

Cost item	Average cost production costs, 2018 (KES)	
	Large-scale farmer	Small-scale farmer
Revenue		
Yield (bags per acre)	24.00	18.00
Sale price (per bag), Jan 2019 ^a	1,867.00	1,867.00
Gross revenue	44,808.00	33,606.00
Costs (per acre)		
Seed	1,949.00	1,620.00
Fertiliser	5,552.00	4,831.00
Pesticides & fungicides	1,406.00	503.00
Herbicides	3,000.00	450.00
Machinery	8,592.00	5,288.00
Labour	6,419.00	8,938.00
Transport (crop output)	1,450.00	-
Others (sisal twines, gunny bags)	1,810.00	1,140.00
Working capital	2,112.00	1,822.00
Production costs	32,290.00	24,592.00
Profit/Loss	12,518.00	9,014.00

Source: (Tegemeo Institute for Agricultural Policy and Development, 2018); EAGC RATIN (for maize sale prices)

The three biggest cost drivers for maize production irrespective of production system are fertilisers, machinery and labour (Figure 11). Collectively, these costs account for 64% of total production costs in large-scale systems and 77% in smallholder production systems. However, labour is a far bigger cost driver for smallholder farmers (36% of total production costs) compared to large-scale farmers, most likely due to limited mechanisation in smallholder farming systems. Conversely, machinery costs are higher for large scale farmers due to higher levels of mechanisation. Fertiliser and seed costs are more-less similar for both production systems, albeit slightly lower cost-share for large-scale farmers compared to smallholders. This similarity is perhaps due to both types of farmers having similar access to these inputs, however, large-scale farmers most likely enjoy economies of scale when purchasing the inputs.

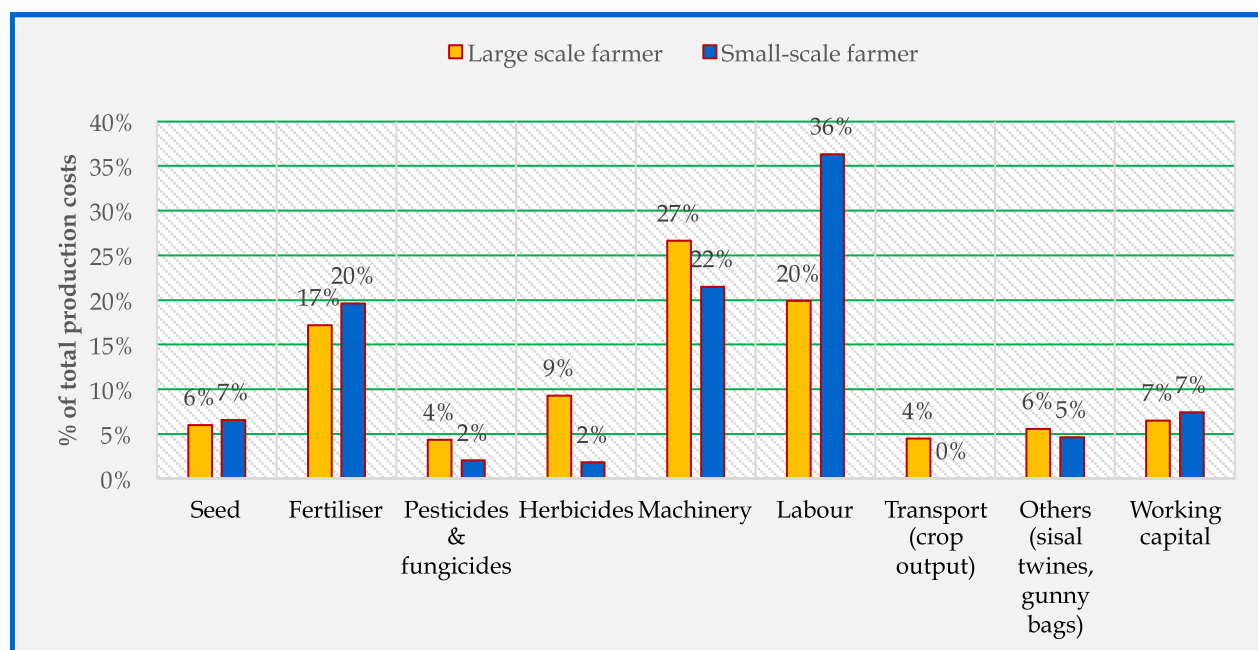
⁵ Tegemeo (2018) define a large-scale farmer as one who farms 50 or more acres of maize, while a smallholder farmer cultivates less than 10 acres of maize. Both types of farmers have 1 season of maize

^a The sale price is for producer markets, proxied by wholesale prices in Eldoret in January 2019 as availed by the EAGC RATIN database



There is significant variation in the use of herbicides between the two systems with large-scale farmers using much more (9% of total production costs) than smallholders (2% of production costs). Furthermore, large-scale farmers incur transport costs for their produce while smallholders do not; this is perhaps because large-scale farmers have more produce and are thus likely to aggregate the same in a storage facility, while for smallholder farmers they are likely to sell produce at the farm gate or use own labour to move the crop to their households or storage facilities.

Figure 11: Comparison of production costs: Large vs smallholder farmers



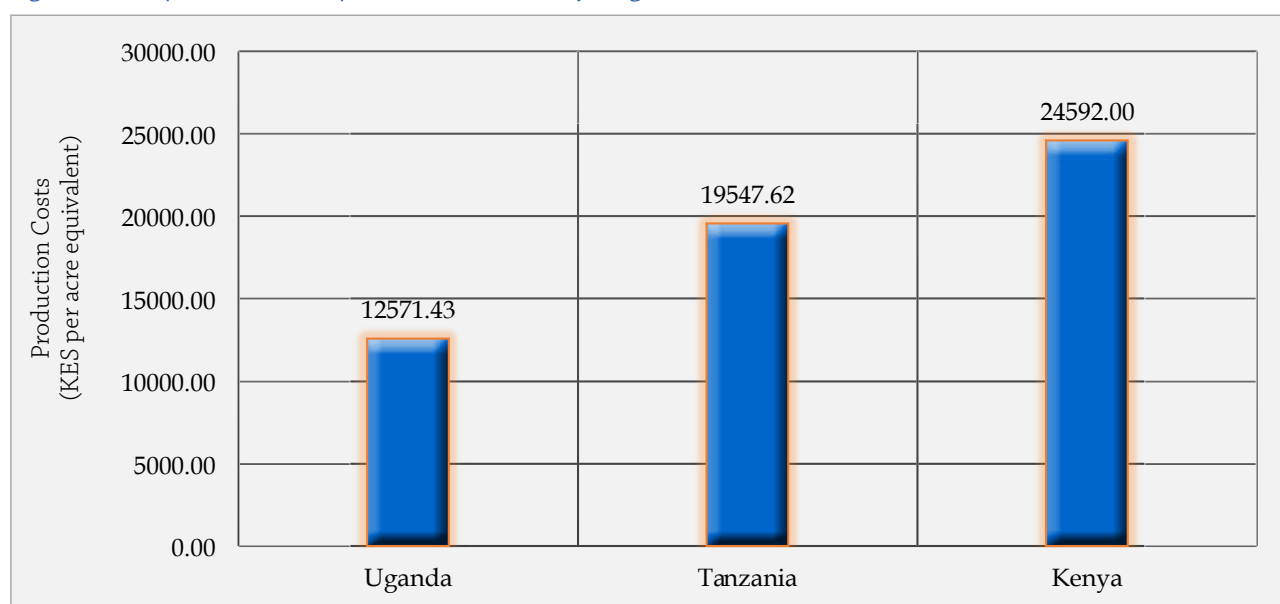
Source: Tegemeo Institute, (2018)

Comparison of production costs with Uganda and Tanzania

Figure 12 shows the comparison between the production of maize in Kenya, Tanzania and Uganda, based on the average costs of production from different farming systems. The results illustrate that Kenya has a higher production cost of maize compared to its neighbours Tanzania and Uganda. A Ugandan farmer is estimated to incur approximately half the cost of his Kenyan counterpart to cultivate one acre of maize. Possible factors explaining Kenya's seemingly high cost of production are that the hybrids used in Kenya require greater use of inputs than the open-pollinated maize varieties used in Uganda. Ugandan production systems have higher soil fertility and favourable weather conditions, conducive to maize production as compared to Kenya (Kiriimi and Swinton, 2004).



Figure 12: Comparison of maize production costs in Kenya, Uganda and Tanzania



Source: EAGC computations, 2020

3.1.2. Challenges Facing Maize Production

There are a diverse array of production challenges and barriers in the maize sector. However, four fundamental constraints impede production. They include the following:

- High input costs: Kenyan farmers are buying and paying more from fertiliser yet, they do not have a comparable increase in yields. The cost of fertiliser is substantially higher in Kenya than in countries such as India. A 50kg bag of Diammonium Phosphate (DAP) fertiliser in Kenya costs between KES 3,200 and 3,500, while the same costs equivalent of KES 2,125 in India⁶.
- Limited mechanization: Smallholder farming is more dominant where family labour is used compared to the large-scale production where mechanisation is viable.
- Declining soil fertility: As indicated in Figure 19, yields have remained relatively stagnant over the past decade due to declining soil fertility caused by poor soil health management. Farmers have repeatedly applied phosphorus-based fertilisers, thus leading to soil acidification. This, coupled with monocropping, has resulted in declining soil fertility.
- Weather variability: The impact of climate change is already being felt, mainly through erratic weather patterns and increased pest and disease outbreak. In 2011, Kenya experienced an outbreak of the maize lethal necrosis disease⁷, and the fall armyworm in September 2016⁸.

3.2. Aggregation and Trade

This segment of the value chain involves the aggregation of grain harvest in different establishments namely: Warehouses, aggregation centers and local storage centres. The major aggregators include small- and medium-sized traders and wholesalers. This node has control of the value chain vested in the knowledge of prices and availability, aggregation and buyers.

⁶ India price estimated from Argus® market report of September 2018. Retrieved from <https://www.argusmedia.com/-/media/Files/sample-reports/argus-phosphates.ashx?la=en&hash=8A9471D217C663184586DCB7B5B05ED435D9DA00>

⁷ See IPPC summary of MLND in Kenya, available here: <https://www.ippc.int/en/countries/kenya/pestreports/2014/06/status-of-maize-lethal-necrosis-disease-mlndin-kenya/>

⁸ See AFA brief on FAW, available here: <https://www.agricultureauthority.go.ke/wp-content/uploads/2018/04/Army-Worm-FAW-Infestation-in-Kenya.pdf>



Small-Scale and Medium Sized Traders

These small-scale and medium sized traders accumulate maize along the value chain. They form the first commercial point in the maize value chain. The number of these traders has been on the rise, penetrating into the villages, thus creating competition along the value chain. The main characteristics of these traders are that they purchase maize within the same district, purchase low volumes, sell immediately, and operate during harvest times. The reasons why most traders do not store maize include liquidity constraints and uncertainty in future prices. They amass the maize in bulk and sell it to wholesalers, and they are not interested in moisture content, since they do not have to store the maize (Kirimi et al., 2011).

Wholesaler Traders

Wholesale traders are important in the maize value chain as they provide working capital to the small-scale traders by paying cash for the maize they purchase. They acquire their maize from small-scale traders or directly from the farmers across a wide geographical area. The majority of these traders have permanent operation centres in major maize surplus towns. In addition, they have permanent centres in the major urban towns, which are the main deficit regions. These traders' own trucks or rent trucks for transportation of their maize across districts. They mainly sell their maize to millers, to the NCPB and directly to consumers in deficit regions. Over half of the major wholesaler's own storage facilities, while the rest rent storage space (Kirimi et al., 2011). The challenge in storing maize for longer periods of time is unpredictable government policies (such as duty waivers and government releases of stock into the market) that may result in losses to the traders. Accordingly, most wholesale traders prefer to store maize for shorter periods.

3.2.1. Post-Harvest Handling, Storage and Management

This segment of the value chain involves the aggregation of grain harvest in different establishments, namely: Warehouses, aggregation centres and local storage centres. This segment is more prominent in markets that do not rely on large-scale modern production. The major aggregators include producer cooperatives, small- and medium-sized traders and processors that have vertically integrated into this stage of the chain (da Silva et al., 2009). Aggregation occurs through multiple layers of small traders, who sell to small-scale processors or exporters. Aggregation mainly occurs to achieve economies of scale so that farmers can satisfy large procurements by processors and thus get value for their produce.

Post-Harvest Losses

Post-harvest losses impact producers and consumers negatively. They reduce farmers' incomes and raise consumer prices as a result of diminished supply. In Kenya, post-harvest losses associated with maize grain are estimated at 12% – 20% of the total national production (Tegemeo Institute of Agricultural Policy and Development, 2017). Some estimates put post-harvest losses as high as 30% (KALRO). The losses mainly include spillages during handling, transportation, processing and marketing; rotting and aflatoxin contamination due to improper handling and inadequate/inappropriate storage technologies; losses to pests such as birds, insects and rodents; and, mechanical damages during farm level elementary processing and off-farm value addition. Together, these account for a loss of between 4.8 million and 12 million bags annually, assuming an average annual production of 40 million bags. In monetary terms, these losses range between KES 14.4 billion and KES 36 billion worth of food lost annually, considering a price of KES 3,000 per bag. This loss would be enough to cover 1.4 months of consumption demand for the country or an equivalent of a complete short-rains harvest, estimated at an average of 5 million bags (ibid).

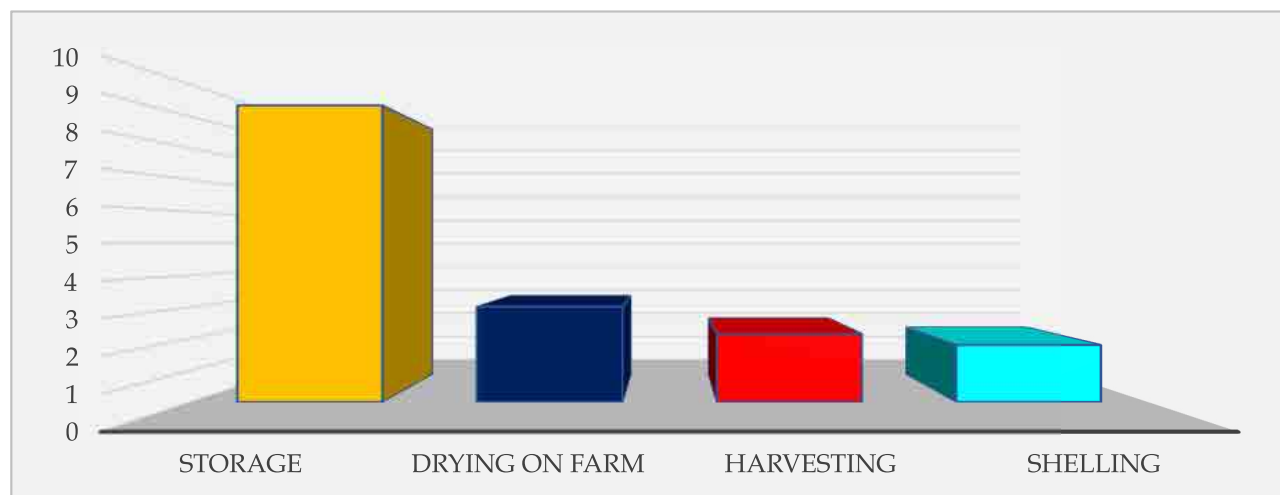
Causes of Post-Harvest Losses

Climate variability and change emerged strongly as a contributor to post-harvest losses in maize. The changing climatic patterns have altered the maize harvesting period conditions in most parts of Kenya's grain basket, with harvesting coinciding with rains or humid conditions in October-November-December. These conditions hinder not only proper grain drying but they also provide conducive conditions for grain pests and disease build-up. Farmers without appropriate storage facilities (a majority of smallholders) suffer huge losses as their produce is either rained on, lost to rot or severely infested by pests and diseases. This is why farmers sell their produce immediately after harvest (Tegemeo Institute of Agricultural Policy and Development, 2017).



In terms of the value chain structure, most losses happen at storage (9% of total production) due to pest infestation, moulding caused by inadequate drying, rodents and poor hygiene (Figure 13). Smaller amounts are also lost during shelling, drying and harvesting.

Figure 13: Causes of post-harvest losses



Source: Computed from FAO (2014)

3.3. Processing

Although wet maize milling to make cooking oil occurs in Kenya, the most predominant form of processing is dry maize milling to make maize meal, flour and maize grits. Other products are oil and feeds.

Hammer Mills (Posho Millers)

These are mainly located in villages in rural areas and they specialize in custom milling whereby the customer provides the grain. Some posho millers had invested in dehullers to produce a more refined product that is most often referred to as “Number 1”. They offer milling services to the farmers for producing un-sifted maize meal. The availability of diesel-operated hammer mills ensures that their services can be made available even in areas where there is no electricity. The expansion of the rural electrification programme has boosted these local mills in rural areas, as most rural centres are connected to the national grid. The milling capacity of these millers ranges between 6 and 110 metric ton per month, but utilization is estimated at about 15% (KAVES-USAID 2014).

The main challenge facing these small-scale millers is limited capital. The market niche that these millers have in the rural areas might be further expanded through assistance to invest in de-haulers. These would enable these millers to produce and package the sifted maize meal in smaller packages, which large-scale millers do not offer. The other constraint faced by the hammer millers is underutilization of the fixed costs, including machines and labour. This is mainly attributed to increased competition, coupled with an expansion of more efficient, electric-powered mills.

Small-Scale Millers

KAVES-USAID (2014) estimate that there are about 50–75 small-scale millers in the country with an installed capacity ranging from 10 to 200 metric tons per month. These millers produce sifted maize meals and maize by-products such as the composite maize meal that they provide in their own consumer brand packaging. These millers operate only in specific towns around the country.

The major constraints faced by the small-scale millers is the use of poor technologies which affects the quality of their products. The other constraints are underutilized processing capacities, limited access to finance, unreliability of grain supply and high cost of power.



Medium and Large-Scale Millers

Medium and large-scale millers in Kenya buy maize at their mill gate from a handful of agents, large-scale farmers, and other direct purchases. They rarely develop buying networks in rural areas, though some will set up satellite depots in major surplus regions. They account for about 93% of the annual total installed milling capacity, estimated at 1.4 million metric tons. Most of the mills are located in Nairobi and Mombasa. Both cities account for 60% of the national installed capacity, followed by Eldoret with 15%, then Thika, Kisumu and Kitale with the remaining installed capacity (KAVESUSAID, 2014). The four main large-scale millers (Mombasa, Unga, Pembe and Eldoret) account for 70% of total milling capacity. The millers produce sifted maize meals in several packages (1kg, 2kg, 5kg, 10kg and 50kg). Apart from sifted maize meal, these millers are also involved in animal feed production.

Some of the medium and large-scale millers own very little grain storage, with most of their warehouse and storage facilities being rented from either private owners or NCPB. Reliance on rented facilities adds additional costs to their operations. Additionally, limited storage, along with some capital constraints, prevent such millers from acquiring all of their annual maize requirements directly. Even in normal years, they depend on NCPB to make up for any shortfalls. However, some millers do have sufficient storage capacity to meet their annual milling requirements. Power shortage and large demand fluctuations are also seen as major constraints to the business of this group of millers.

3.4. Distribution to Consumers

These players include urban and rural consumers. Urban consumers source their maize meal from supermarkets, wholesalers and retailers. Rural consumers source their maize meal mainly from retailers (dukas/kiosks). In the urban setting, the packaging of maize meal is usually in 1-, 2-, 5-, 10- and 50-kilogram parcels, while in the rural areas, packaging is mainly in 1-, 2- and 5-kilogram parcels. Retailers in rural areas prefer purchasing maize meal in 50-kg bags that they parcel out and re-sell to rural consumers in smaller quantities of their choice (KAVESUSAID, 2014).



4. Maize Marketing and Trade



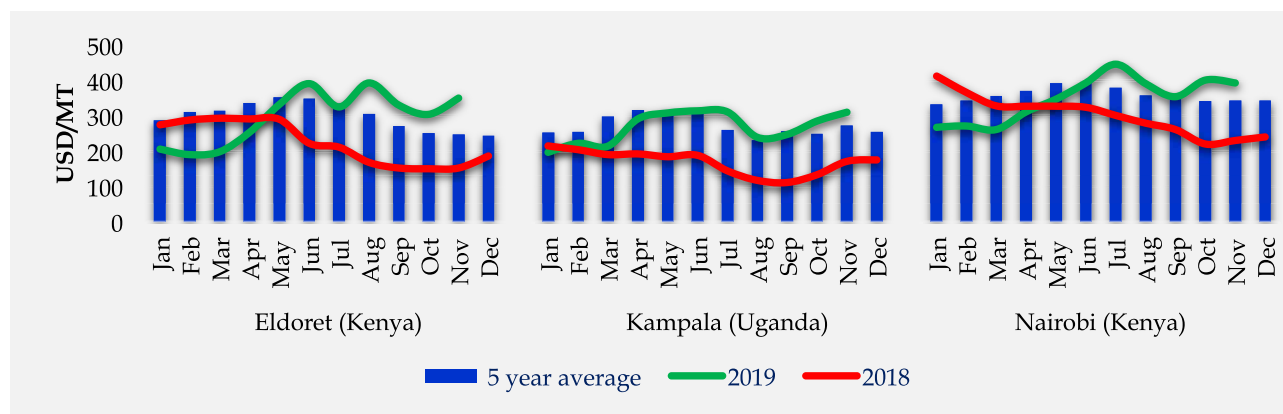
4 Maize Marketing and Trade

4.1. Maize Price Trends

Maize prices change seasonally depending mainly on relative supply to demand. Occasionally, policy decisions by the government also influence market forces and prices. The amount of maize is primarily determined by the local harvest; the primary harvesting season is the fourth quarter of the year (October to December); consequently, prices are usually lowest during this period. On the other hand, prices are generally highest during the second quarter of the year which is the lean period; during this period, prices in Nairobi, the primary consumer market, are on average 15% higher than during the harvest season (**Figure 14**).

Maize prices in producer regions (proxied by Eldoret in Figure 14) generally follow similar patterns to those in consumption regions, albeit with more volatility. For instance, prices during the peak of the lean period are up to 44 % higher than during the peak of the harvest season. This increased volatility is because producers (and by extension production regions) are price-takers, with consumption regions such as Nairobi being price-makers. Maize prices in 2019 were generally higher than in 2018 and the 5-year average due to lower domestic availability, delayed harvests in neighbouring countries. Reduced export volumes to Kenya from Tanzania occasioned by high demand for maize from Southern Africa (EAGC, 2019).

Figure 14: Maize prices in selected markets (2018-2019)



Source: EAGC RATIN, 2019



Considering that Kenya is a net importer of maize, Nairobi is an important reference market for maize prices in East Africa. Price movements in Nairobi influence prices in Uganda and Tanzania, particularly as the harvests in the latter two countries coincide with the lean season in Kenya. An analysis of maize prices in East Africa (Table 4 below) indicates a strong positive correlation between maize prices in Nairobi with those in Kampala, Arusha and Busia, in addition to expected correlation with Eldoret prices⁹.

Table 4: Correlation analysis of maize prices in selected East African markets, 2014-2019

	Nairobi	Eldoret	Arusha	Kampala	Busia
Nairobi	1				
Eldoret	0.823797	1			
Arusha	0.723113	0.744814	1		
Kampala	0.66712	0.765694	0.788296	1	
Busia	0.685305	0.820156	0.807362	0.976142	1

Source: Computations from EAGC RATIN, 2019

4.1.1. Import Parity Analysis and Implications for Buying Decisions by Millers

The price of maize domestically compared to neighbouring countries is perhaps the most important consideration when millers make decisions on whether to purchase maize locally or import from neighbouring countries. This is illustrated by the import parity price (IPP) estimations, which approximates the price of imported commodities in the importing country and indicates the viability of the imports (in this case, a miller). It includes the cost of purchase of an imported commodity, applicable import duties and taxes, administrative costs of imports and transport to the importers' location. If the IPP is lower than domestic prices, it makes financial sense to import; conversely, if the IPP is higher, it is more prudent to purchase the commodity from domestic sources, other factors remaining constant.

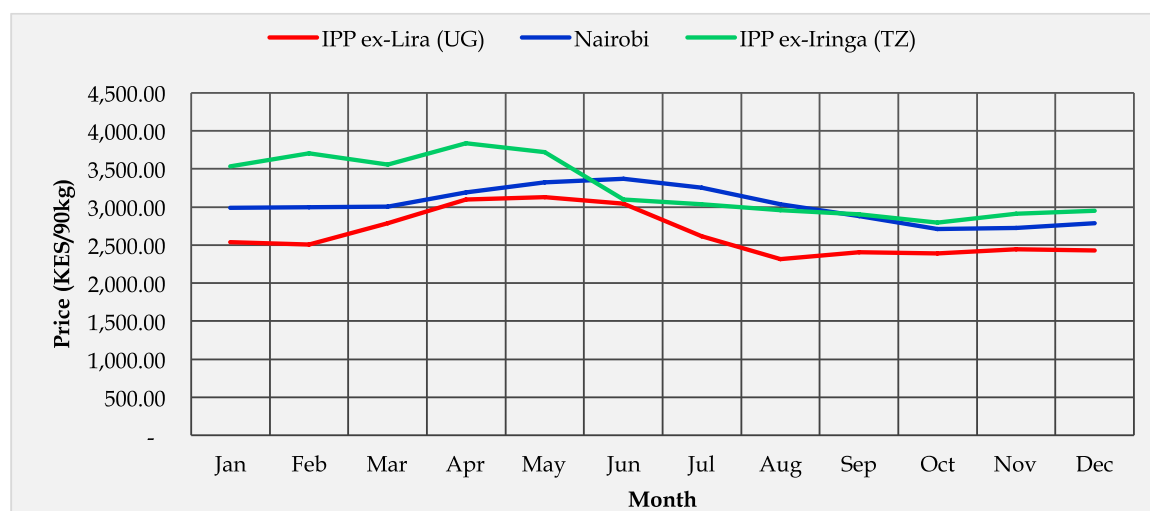
An analysis of five-year average import parity prices for maize from Lira in Uganda compared with those from Iringa in Tanzania (both maize basket regions in their respective countries), reveals that it is cheaper to import maize from Uganda to Kenya throughout the year, all other factors remaining constant (**Figure 15**). This is the case during the third quarter of the year when there is a considerable divergence between IPP ex-Lira and Nairobi prices, caused by increased supply from Uganda.

On the other hand, maize imports from Tanzania make business sense between June and September but are impractical during other months. Like Uganda, the June to September period also coincides with Tanzania's primary harvest season. These trends explain the propensity of millers to import maize from Uganda, despite the quality concerns compared to Tanzania.

⁹ Correlation measures the degree of relationship between variables. It is measured between 0 and 1 where 1 indicates perfect correlation while 0 indicates no correlation. Negative and positive signs indicate the nature of the correlation, either inverse or positive correlation respectively.



Figure 15: Import parity, Lira (UG) vs Nairobi (KE) vs Iringa (TZ), 5-yr average prices, 2014-2018

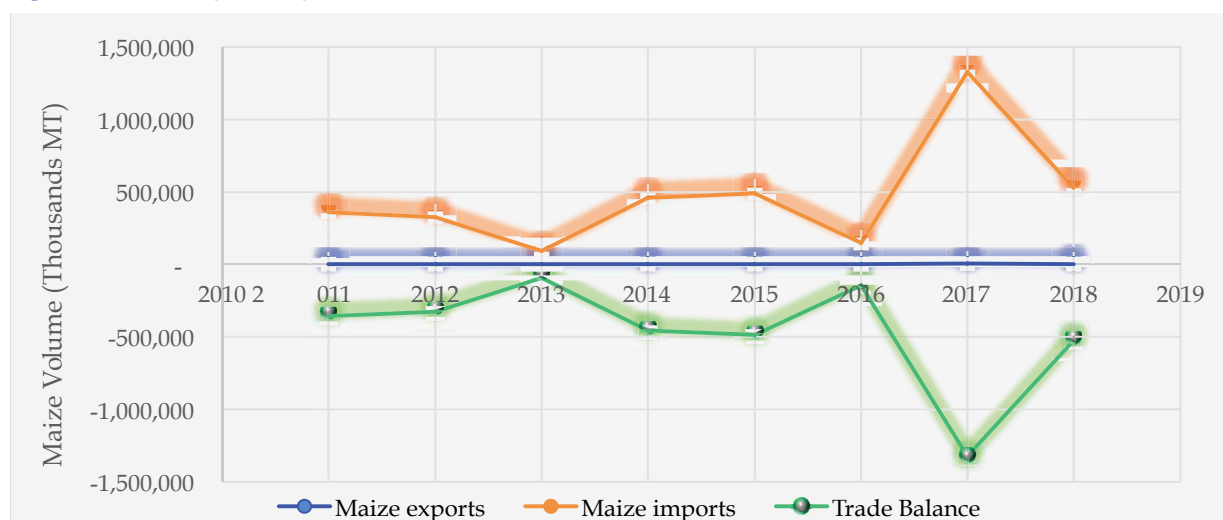


Source: Computed from EAGC RATIN and EAGC Logistics Calculator

4.2. Cross-Border Trade Patterns

Kenya is a net importer of maize due to higher consumption relative to domestic production. Additionally, other countries in Eastern Africa such as Uganda, Tanzania and Zambia have more favourable agro-ecologies and lower production costs for maize, thus giving them a comparative advantage over Kenya in maize production. These factors, coupled with increased consumption as the population grows, has led to a consistent increase in imports and trade deficit in maize since the early 2010s. As illustrated by **Figure 16**, maize imports have increased by 47% between 2011 and 2018, with peak imports in 2017 during the regional maize shortage that resulted in an import surge from international markets. The maize trade deficit increased to over 526,000MT in 2018 from 358,000MT in 2011.

Figure 16: Maize imports, exports and trade balance, 2011-2018

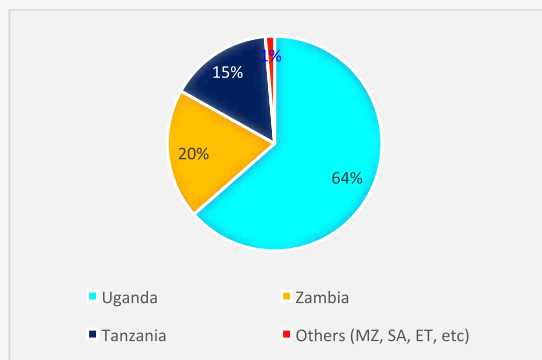


Source: KNBS Economic Survey 2016 and 2018

Most of the maize imported into Kenya originates from Eastern Africa, with Uganda accounting for approximately 64% of maize imports into Kenya (**Figure 17**). Zambia and Tanzania respectively account for 20% and 15% of imports into Kenya, with relatively minimal imports. Interestingly, the composition of the main trade partners for maize has changed over the past five years. In 2014, Tanzania accounted for 71% of maize imports by Kenya with Uganda and Zambia accounting, for 16% and 11% respectively (**Figure 18**). The fact that Uganda has replaced Tanzania as the main exporter to Kenya is most likely due to propensity to impose export bans on maize as was the case in 2016 and 2017, which even when lifted, leave an air of uncertainty amongst traders and millers. Uganda, on the other hand, has a much more open-trade policy and has no history of export bans, even during periods of depressed harvests.

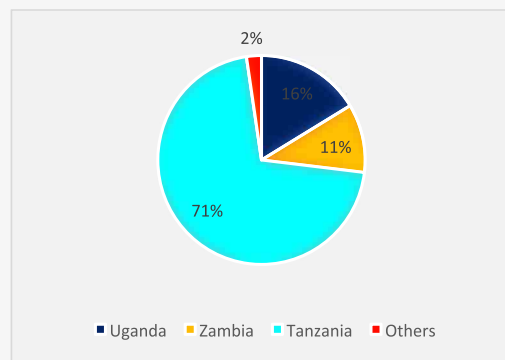


Figure 17: Composition of maize imports, 2018 (MT)



Source: ITC Trade Maps

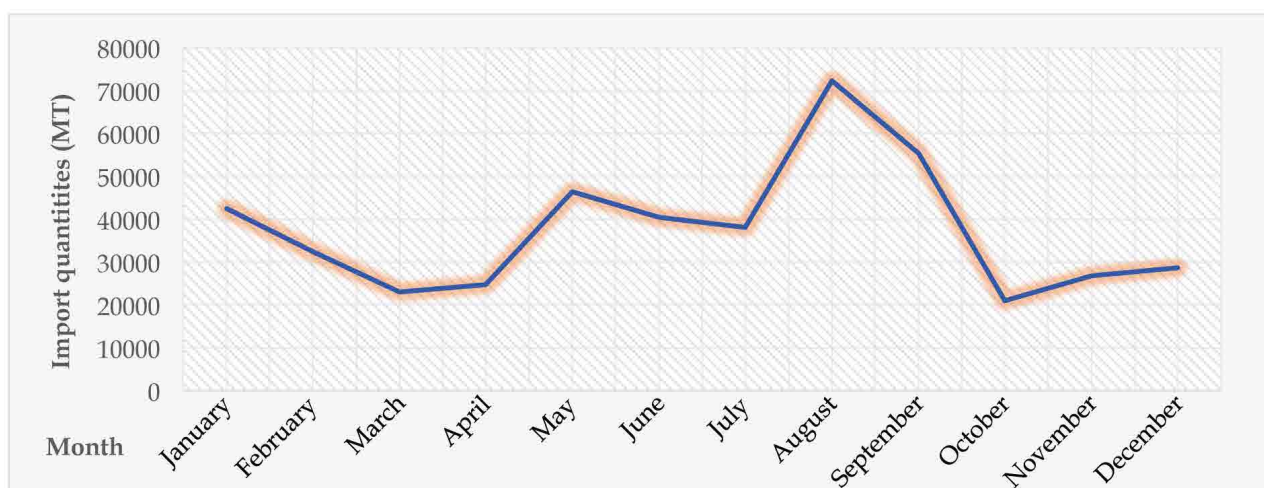
Figure 18: Maize imports, 2014 (USD thousands)



Source: ITC Trade Maps

Cross-border trade flows are aligned to the respective crop calendars for Kenya and its main trading partners (Uganda and Tanzania). Peak imports are usually during the second and third quarters of the year, which coincide with the lean period domestically and harvests from trading partners. Imports from Uganda typically peak in the third quarter while those from Tanzania peak during the second quarter. Conversely, import volumes are much lower during the first and fourth quarters due to relatively high domestic availability of maize from the harvest season which takes place between October and December. Figure 19 below, based on data on informal cross-border trade, illustrates these trends more clearly.

Figure 19: Five-year average monthly imports from Uganda and Tanzania, 2014-2018



Source: Computed from EAGC RATIN and FEWSNET Data

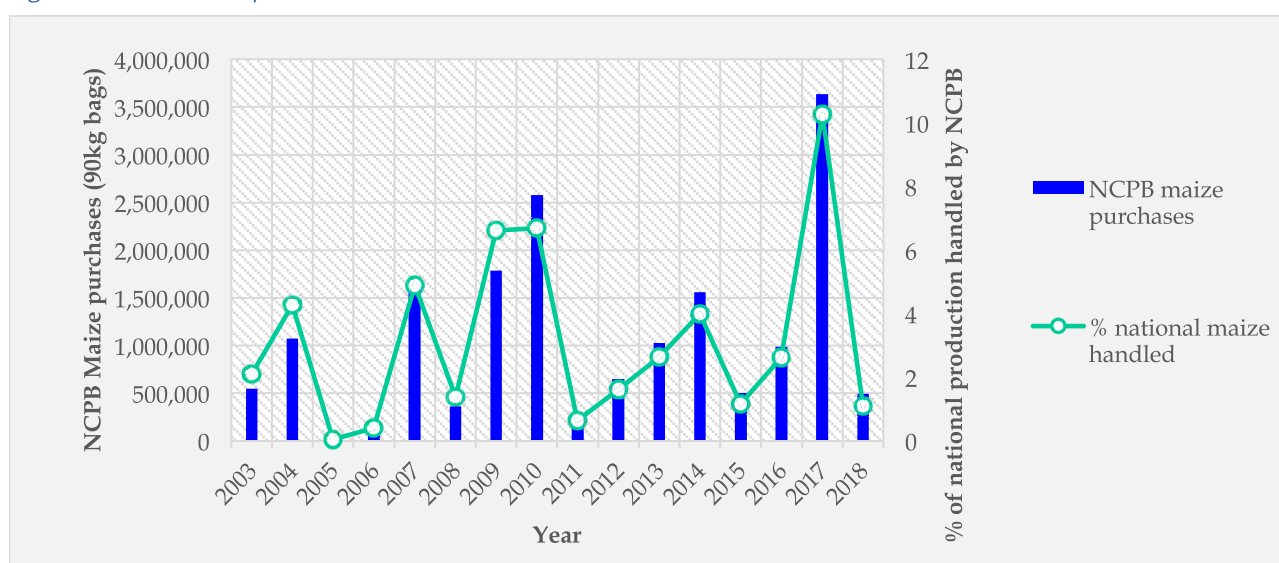


4.3. Domestic Trade Dynamics

Smallholder farmers retain approximately 58% of their produce for household consumption (Onono, Wawire, & Ombuki, 2013). Small and Medium Enterprises (SME), larger traders and millers usually purchase the marketed surplus.

The National Cereals and Produce Board (NCPB) participates in the maize market, predominantly as an agent of the government in maintaining the Strategic Food Reserve (SFR), but also as a commercial entity as per its Founding Act. The volume of maize purchased or transacted by NCPB is relatively small, averaging approximately 3% between 2003 and 2017 (**Figure 20**) (NCPB, 2018). The highest volume handled by NCPB in recent times was in 2017 when it handled over 3.6 million bags. Most of this maize was imported duty-free from outside Eastern Africa and sold to millers at subsidised prices due to shortages locally and regionally.

Figure 20: NCPB maize purchases, 2003-2018



Source: NCPB (2018)

NCPB typically purchases maize on behalf of the government at above-market prices. Therefore, despite the relatively small volumes that it handles annually, its impact on domestic prices is quite pronounced because it automatically becomes the buyer of first resort for farmers and traders, forcing millers to match NCPB prices to obtain supplies.

4.3.1. Government participation in the Maize Value Chain: The Case of NCPB and Strategic Food Reserve Trust Fund (SFRTF)

The government typically plays an active role in maize markets with the aim of enhancing food security, price stabilisation and/or facilitating market access for farmers. The NCPB and the Strategic Food Reserve Trust Fund (SFRTF) are the two leading public organisations through which the government interacts directly with maize markets. This section profiles the two organisations and their role and impact on the maize value chain in Kenya.



a) The National Cereals and Produce Board

NCPB was established by the National Cereals and Produce Board Act, Cap 338 of 1985 as a body corporate with a mandate to regulate and control the marketing and processing of maize, wheat and scheduled agricultural produce and related purposes. Section 4 of the Act presents the key functions of the Board as:

- i. To regulate or to control the collection, movement, storage, sale, purchase, transportation, marketing, processing, distribution, importation, exportation, disposal and supply of maize, wheat and scheduled agricultural produce;
- ii. To buy, store, sell, import, export or otherwise acquire and dispose of maize, wheat and scheduled agricultural produce in such manner, such quantities and on such terms as it may, from time to time, deem necessary to fulfil the requirements of producers and consumers in Kenya;
- iii. To advise the Minister on the proper production of maize, wheat and scheduled agricultural produce in relation to the needs of Kenya, and the extent to which control over the exportation and importation of maize, wheat or scheduled agricultural produce is desirable or necessary; and
- iv. To do any other act which is connected or incidental to the foregoing.

Before the Act, NCPB traces its origins to the Kenya Maize Board established in 1942 to market European maize and the Maize and Produce Control Board to market African maize and other crops (NCPB, 2018). Post-independence, the Maize & Produce Marketing Board was established in 1967 by merging the former marketing boards, except the wheat marketing board (Mutai, 2012). Nonetheless, in 1979, the Maize & Produce Board and the Wheat Board of Kenya were merged to form NCPB to consolidate control and marketing activities of maize, wheat and other commodities (NCPB, 2018).

NCPB was commercialised during the implementation of the Structural Adjustment Programmes and was expected to generate revenue from commercial activities such as commodity trading, leasing out surplus storage facilities and provision of agency services for the Government of Kenya for handling grain for the Strategic Grain Reserve (now transformed into a Strategic Food Reserve) (NCPB, 2018).

Storage Facilities

NCPB currently has 110 storage facilities with a combined storage capacity of approximately 1.8 million MT (20.45 million 90kg bags). NCPB also owns Cyprus bins, which are semi-underground storage facilities that allow hermetic (air-tight) storage of grains. However, these are no longer in use.

NCPB warehouses are available in 46 of the 47 counties, with the exception being Tharaka Nithi. The location of these warehouses is presented in **Table 5 below**, clustered in NCPB operating regions. Most of the storage capacities are concentrated in the key production regions in the Rift Valley and Nairobi as the key consumption hub.

Table 5: Capacity of NCPB storage per location

Location (NCPB Region)	Total capacity (MT)	Total capacity (90kg bags)
Nairobi/Eastern	338,845	3,764,944
North Rift	394,020	4,378,000
South Rift	438,300	4,870,000
Western	199,890	2,221,000
Nyanza	156,330	1,737,000
Coast	87,570	973,000
Northern	225,360	2,504,000
TOTAL	1,840,315	20,447,944

Source: NCPB, 2019



However, there have been concerns over the relatively poor state of NCPB warehouses to guarantee the quality and quantity of commodities stored therein. As such, the government through the State Department of Trade intends to invest KES 2.6 billion to upgrade some of the NCPB facilities to meet requirements for storage of agricultural commodities for trade on the Kenya Commodity Exchange through the Warehouse Receipt System¹⁰.

Commercial Services

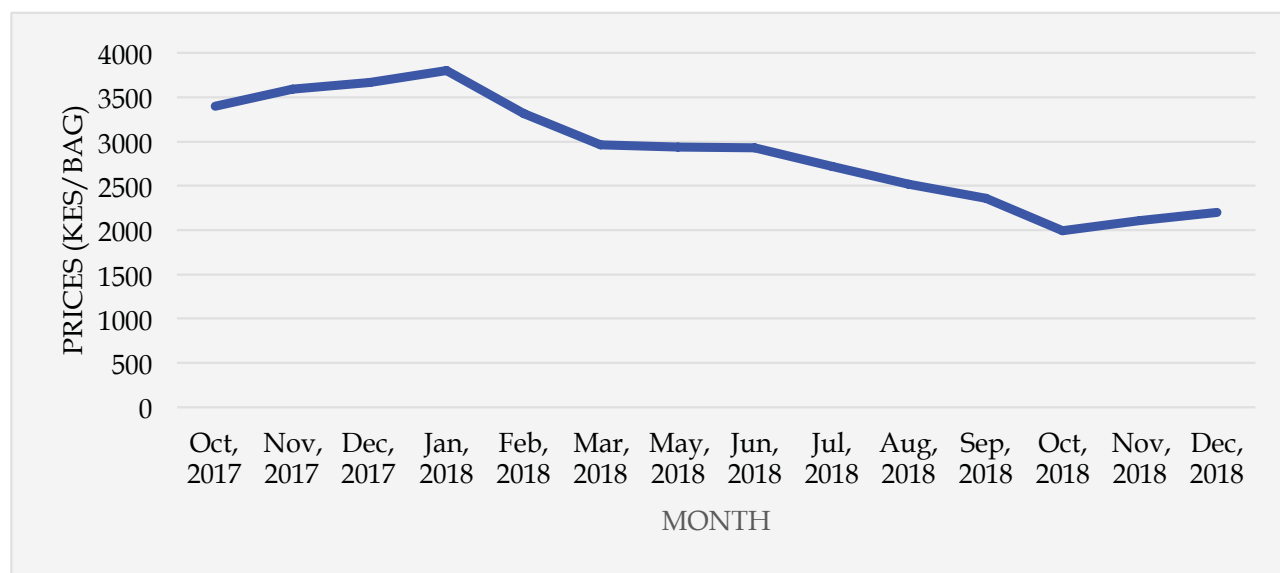
- i. Grain quality and conditioning which includes drying, cleaning, grading and quality tests
- ii. **Grain storage services** that cover bagging, warehousing and fumigation
- iii. **Logistics**, which include transport, distribution, clearing and forwarding services
- iv. **Value addition/milling services**

Challenges in NCPB Participation in Maize Markets

Despite its noble intentions and the relatively small volume of maize handled by NCPB on behalf of the government, three main challenges have arisen as described below:

- i. **Creates price uncertainty:** During harvest, maize prices become artificially inflated, particularly at the time of harvest because NCPB/SFRTF become buyers of first resort for farmers. For instance, during the harvest season in 2017, the government announced that it would purchase maize from farmers at KES 3,200 per 90kg bag through SFRTF and its agent, NCPB. At the same time, the average maize price in Eldoret and Nakuru was between KES 2,600 and 2,800 per 90kg. Subsequently, prices in Nairobi increased by 12% between October 2017 and January 2018. With limited capacity by NCPB to absorb the surplus and millers opting for cheaper Tanzanian and Ugandan maize, the prices declined significantly for the remainder of the year; prices in December 2018 were 42% below January 2018 prices (Figure 21).

Figure 21: Nairobi Maize prices, 2017/18



Source: EAGC RATIN, 2019

¹⁰ Remarks by Dr Chris Kiptoo, Principal Secretary in the State Department for Trade during a workshop on the WRS and Commodity Market Regulations with the National Assembly Departmental Committee for Delegated Legislation on 29th November 2019 in Mombasa, Kenya



- ii. **Compromised quality of maize in NCPB depots:** Millers point out that the quality of maize distributed by NCPB is usually of inferior quality and occasionally unfit for human consumption. A report by Kenya Bureau of Standards (KEBS) in 2018 revealed that 63.3% of maize in NCPB silos, worth over KES 7 billion at the time, was unfit for human consumption¹¹. NCPB also acknowledges challenges with the quality of maize in its depots, noting that the Board does not have adequate facilities to monitor grain quality as it has only one laboratory to cater for its facilities scattered in 46 of the 47 counties in the country (EAGC, 2018; NCPB, 2018).
- iii. **Bureaucracies at NCPB:** Millers pointed out that the process of collecting maize stocks from NCPB depots is fraught with bureaucracies through which corrupt practices manifest themselves. Specifically, millers pointed out the following bottlenecks:
 - a. Long queues at NCPB depots to buy maize whereby millers wait up to 3 days to collect maize. During that time, they must cover the costs of accommodation and meals for their staff assigned to receive the maize
 - b. Some NCPB staff allegedly demanding bribes to allow purchases of maize, even if the miller has fulfilled all the requirements.
 - c. Similar to (b), some of the bags sold from the NCPB stores are underweight. In one case, a miller in Nakuru who bought 200 bags of 50kg from NCPB later found that each bag weighed less by between 5kg to 10kg. Assuming a 5kg loss in each bag, this is equivalent to 1MT of maize lost.

b) The Strategic Food Reserve Trust Fund

The Strategic Food Reserve Trust Fund (SFRTF) was established in 2015 as a trust fund -Strategic Food Reserve Trust Fund Regulations, 2015 under the Public finance Management Act 2012. The broad purpose of this fund is to stock and distribute critical foodstuffs from the strategic food reserves during food emergencies. Specifically, the fund is expected to:

- a) Stabilize the food supply and prices in the country;
- b) Arrange for procurement, storage and sale of food commodities;
- c) Maintain adequate strategic food reserves in physical stock or cash equivalent at any one given time; and
- d) Mobilize resources to support strategic food reserve related activities.

Challenges in operating the SFRTF include the limited public availability of explicit buy/sell policy guidelines thus creating uncertainty in the market; lack of adequate emergency cash reserves, this leads to insufficient stock during emergencies. The SFRTF has a mandate that covers six commodities: maize, beans, rice, canned beef, powdered milk and fish. However, due to limited funding previous purchases by SFRTF show that more than 95% of all purchases were of maize through NCPB, with the rest being primarily powdered milk and beans (MoALE, 2019). For this fund to run effectively, the maize taskforce report (2019) recommended the need to develop a Strategic Food Reserve Act and guidelines to operationalise the act. In addition to enhancing the SFR structures to allow the establishment of an autonomous and independent entity.

¹¹ See <https://www.nation.co.ke/news/State-orders-fresh-audit-on-bad-maize-stored/1056-4840204-124w2riz/index.html>



Implications of the Strategic Food Reserve on the Maize Value Chain

An efficient and effective SFR presents the following opportunities for the private sector:

- Participation in the maize procurement processes, thus minimising government disruption in the market. Some progress is already being made in this regard. In 2018/2019, the Government purchased maize from farmers at considerably lower prices compared to previous years (KES 2,500 per bag announced in January 2019 compared to KES 3,200 in October 2017). The maize procurement process has been reformed to allow the vetting of suppliers, thus facilitating traceability of stocks back to the farmer.
- Participate in the provision of storage services: Private sector operators can now compete with NCPB to store grain on behalf of the SFR. This will encourage innovations and efficiency in maize storage for food reserve purposes and will help to reduce post-harvest losses and sale of poor-quality maize to millers during lean periods.



5. Analysis of the Maize Milling Sub-Sector – Small and Medium Scale Millers





5 Analysis of the Maize Milling Sub-Sector – Small and Medium Scale Millers

5.1. Description of the Respondents

The study carried out a survey focusing on maize flour production for human consumption, where 17 millers and two millers' associations were interviewed against a target of 30 millers (response rate of 56%). The spatial distribution of the respondents is shown below:

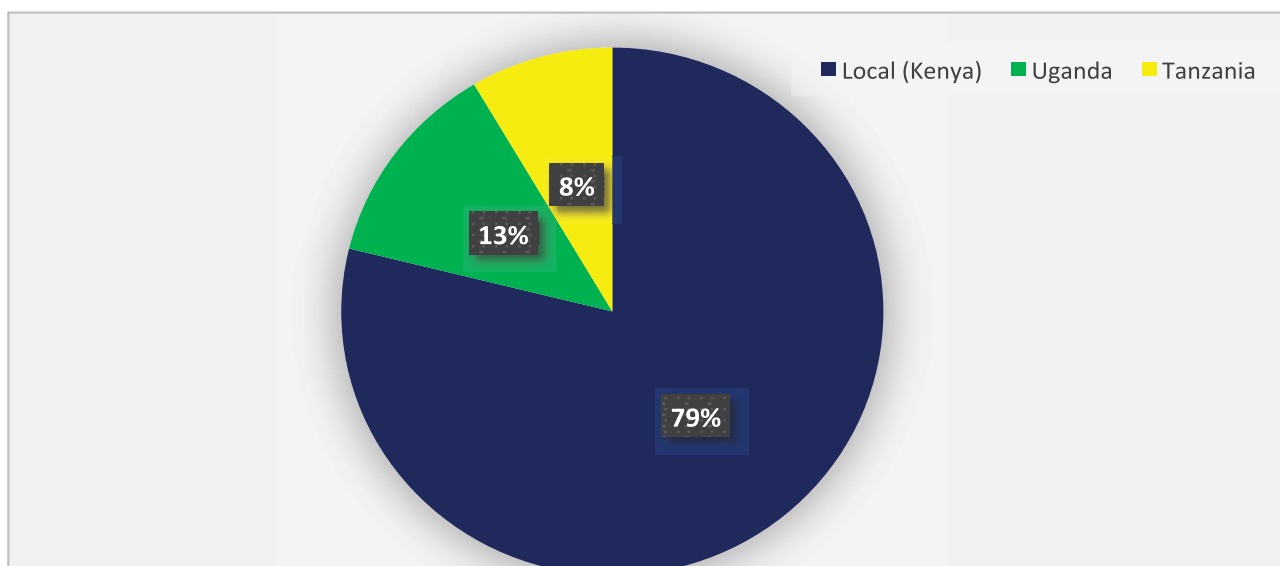
Location	Number of millers
Nairobi	1
Kiambu	6
Muranga	1
Meru	2
Uasin Gishu	2
Trans-Nzoia	1
Bungoma	1
Nakuru	2
Mombasa	1

Sources: Field survey, 2019

The maize milling sub-sector is dominated by small and medium scale millers, who are estimated to control approximately 60% of the market. These millers generally have a milling capacity of not more than 150MT per day. The respondents in the study had an average milling capacity of under 60MT per day. And according to the Cereal Millers Association, large-scale millers account for approximately 40% of the market with an installed milling capacity estimated at 6,600MT per day. The primary source of maize grain was the local market, particularly from the grain basket regions in Central and North Rift counties. On average, the millers obtained approximately 79% of their milling requirements domestically with 21% being imported from neighbouring countries (Uganda and Tanzania), as illustrated by **Figure 22**.



Figure 22: Sources of maize for milling

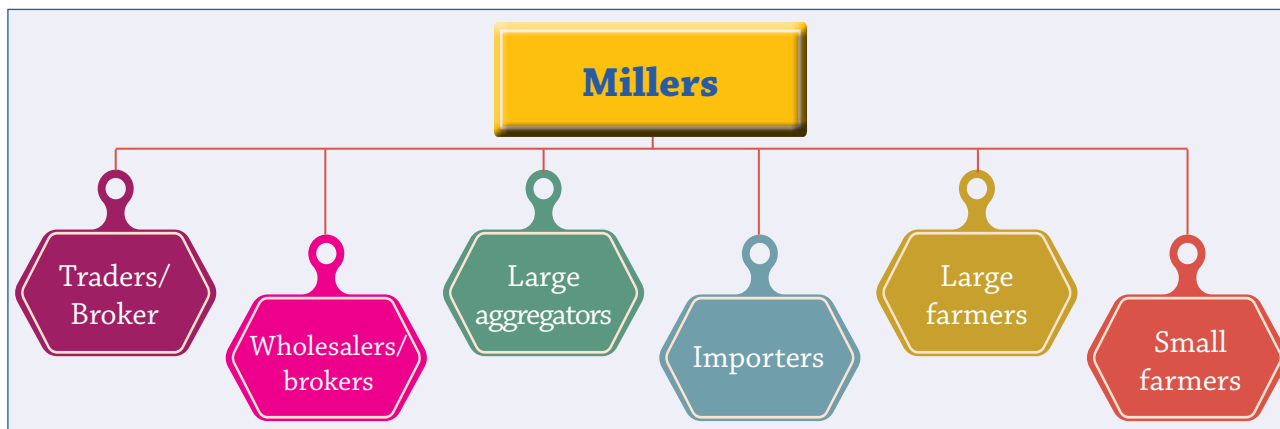


Sources: Field survey, 2019

5.2. The Millers Supply Chain

In the absence of sophisticated, structured grain trading systems, millers in Kenya typically operate in the spot market almost on a day-to-day basis based on the volume of deliveries, their stockholding capacity and available cash or credit lines. In doing so, millers deal with multiple individual traders who deliver maize to their factories daily (Figure 23).

Figure 23: A miller's supply chain



Source: EAGC, 2019

Additionally, deliveries are not standardised; each trader delivers different quantities of maize of varying origins and quality requirements. This means that a miller must manage numerous business relationships for supply contracts, payment, delivery terms and quality requirements. A multiplicity of these relationships.

presents costs to their business, which could have been reduced or avoided altogether by trading through a commodity exchange or trade platform.

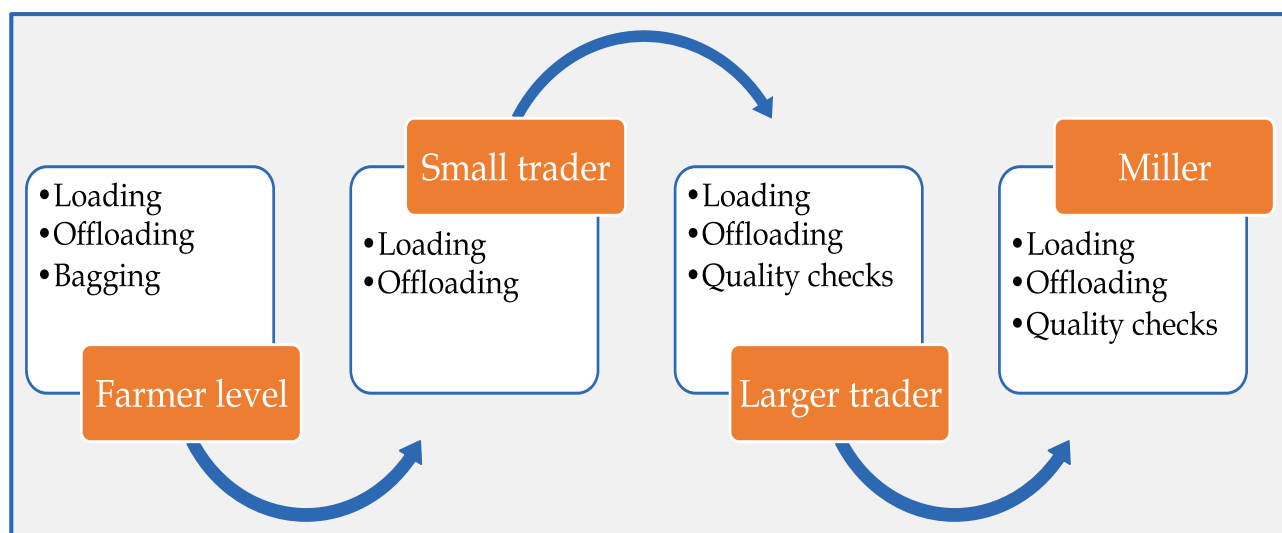
In jurisdictions with well-functioning, structured grain markets such as South Africa, millers purchase maize for milling through commodity exchanges or trade platforms using forward and futures contracts. These contracts allow millers to hedge against price risks several months in advance by taking a position on the market. Deliveries are also guaranteed upon maturity of the contracts, which allows millers to synchronise and optimise their milling operations.



5.2.1. Multiple Levels of Grain Handling and Cost Implications for Millers

As illustrated in the maize value chain map in section 3, there are at least four stages of maize handling up to the milling stage, as shown by Figure 24. At each level, a handling cost is charged on each bag of maize. Loading and offloading costs are standard across all levels; quality checks are more common at trader and miller level. Additionally, there are also costs of bagging at the farmer and/or trader level. Stakeholder interviews revealed that these handling costs could add up to KES 200 to the cost of each bag before milling. This cost is the equivalent of approximately 8% of the price of a bag of maize in Eldoret.

Figure 24: Multiple levels of grain handling from farm to miller



Source: EAGC

Impact of Post-Harvest Losses, Handling and Storage Challenges on Maize Millers

Post-harvest losses, handling and storage challenges at farm level affect millers in three main ways:

- Reduce the volume available for milling: The 2019 season is a case in point; the rains undermined the pace of harvest. Due to this, the maize market remained thin even during the harvest period. This meant that several millers were operating below their capacity. At the time of this survey, most millers indicated that they were operating at 36% capacity.
- Increases the price of maize for millers: After (i) above, maize prices remained high throughout the harvest period. At the time of this study, millers in Nairobi and Thika were paying KES 3,500 per bag for maize deliveries, while the prevailing price in Eldoret was approximately KES 2,700 per bag. By comparison, in the same period in 2018, Nairobi/Thika and Eldoret prices were approximately KES 2,202 and KES 1,707 per bag respectively.
- Compromises the quality of maize: Taking delivery of wet maize increases the risk of aflatoxin contamination and prevalence of rotten and diseased grains. The quality of maize is the leading concern amongst millers, followed by the cost of quality control (laboratory costs).



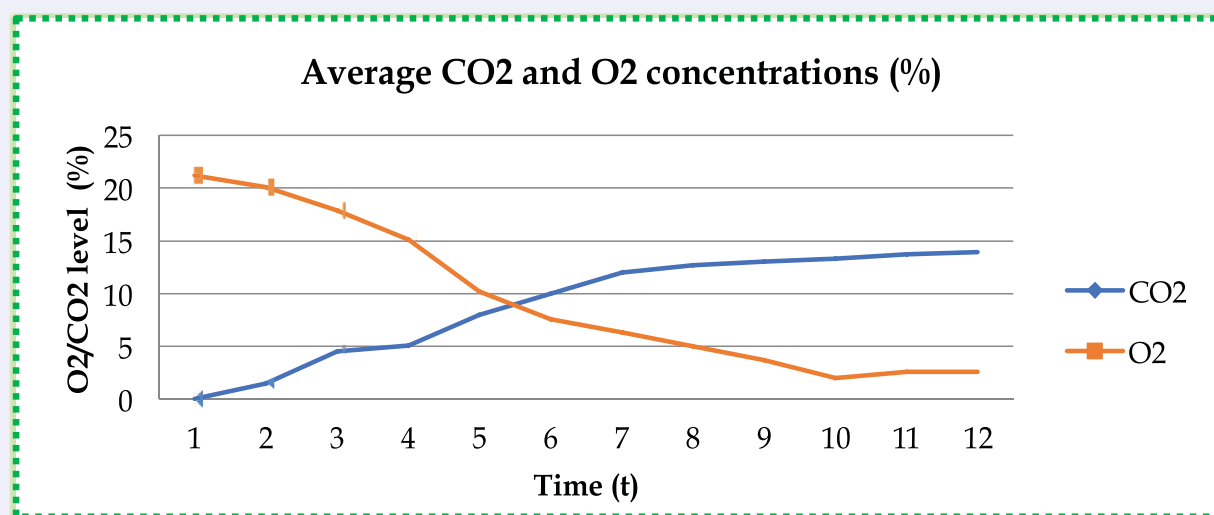
Recent post-harvest handling and storage innovations: The case of hermetic storage technologies

Perhaps the most important innovation in post-harvest handling and storage is the introduction of hermetic storage technologies (HSTs) in the market. Hermetic storage entails the storage of commodities in an airtight container; in the process, oxygen within the container is gradually depleted and replaced by carbon dioxide as metabolic activity of organisms contained therein continues. The depletion of oxygen asphyxiates the organisms and prevents further multiplication (see Figure 25 below). Hermetic storage thus allows the control of grain storage pests without the use of insecticides.

HSTs include:

- Hermetic storage grain bags, which feature an outer woven polypropylene bags and inner hermetic liner(s). These usually store 50-100kg of grain (maize equivalent).
- Metal silos, typically fabricated by artisans
- Rigid plastic grain silos, typically with a storage capacity of 380kg (approximately 4 bags of 90kg)
- Hermetic silo bags, which allow bulk storage of grains, some of which have a capacity of up to 200MT (EAGC, 2018).

Figure 25: How hermetic storage works



Source: EAGC, 2019

EAGC spearheaded the development of the Kenya Standards for the hermetic storage technologies, specifically for hermetic storage bags and the rigid plastic silo, which were gazetted in August 2019. These standards were demanded by manufacturers and end-users of HSTs on account of increased prevalence of sub-standard products in the market purporting to be HSTs, thus undermining manufacturers and efforts to reduce post-harvest losses.

5.2.2. Costs of Handling and Storage for Farmers: A Case Study of Nakuru

Post-harvest handling and storage of grains is a costly affair for maize farmers. Field surveys in Nakuru County during this study indicated that farmers usually harvest the maize at relatively high moisture content, sometimes exceeding 20%, and then sun-dry the grain to the required moisture content of 13.5% as per maize standards. Sun-drying is the most preferred drying method for farmers as it is the less costly (sun-drying costs approximately KES 30 per bag per day with constant sunshine, the desired moisture level can be attained within 3 to 4 days).



However, at the time of this study, the harvest had coincided with heavy and prolonged rainfall which made sun-drying impossible. Farmers thus had two options, either sell their produce wet at low prices or dry their maize mechanically through grain dryers. NCPB offers drying services at KES 39.44 per percentage point reduction in moisture, per bag. This is slightly higher than drying fees charged by private sector operators who charge on average KES 35 per percentage point reduction in moisture per bag.

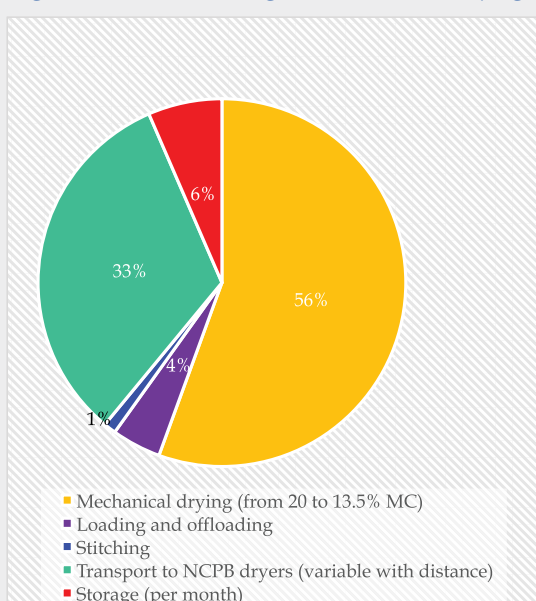
Table 6 summarises the drying, storage and handling costs incurred by farmers in Nakuru. As expected, mechanical drying accounts for a more substantial portion of the post-harvest costs (56%) as opposed to sun-drying, which accounts for 34% (Figure 26 and Figure 27 below). In addition to drying fees, farmers must incur costs of loading and off-loading, stitching and transport from their farms/aggregation points to NCPB locations.

Table 6: Cost of grain drying, storage and handling in Nakuru

Cost item	Unit cost (KES/bag)
Drying per percentage point moisture drop (NCPB dryers)	39.44
Sun-drying (3-4 days in constant sunshine)	90-120
Loading and offloading	20
Stitching	5
Transport to NCPB dryers (variable with distance)	100-200
Storage (per month)	30

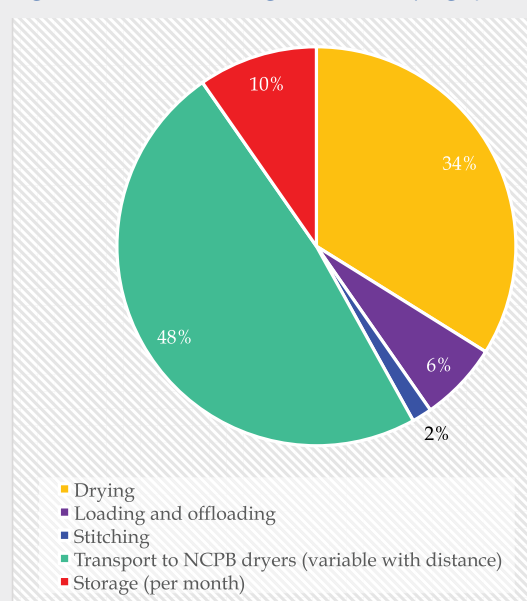
Source: EAGC Field surveys, 2019

Figure 26: Grain handling costs mechanical drying



Source: Field Surveys, 2019

Figure 27: Grain handling costs (sun-drying option)



Source: Field Surveys, 2019

Impact on Farmers' Earnings

The above storage and handling costs mean that a farmer can expect to incur over KES 500 per bag on storage and handling, assuming they harvest maize at around 20% moisture content (**Table 7** below). Considering that smallholder farmers earn approximately the same amount from farm operations (**Table 3**), grain handling and storage costs will most likely wipe out all farmers' earnings. Thus, the propensity of farmers to sell their commodities at farm-gate to small/medium traders because of post-harvest handling and storage operations do not make business sense for most farmers.



Table 7: Comparison of costs, sun vs mechanical drying

Cost item	Total cost (per 90kg bag)	
	Mechanical drying	Sun-drying
Drying	276.08	120
Loading and offloading	20	20
Stitching	5	5
Transport to NCPB dryers (variable with distance)	200	200
Total cost	501.08	345

Source: Field interviews

On the other hand, sun-drying is typically more affordable and thus preferred by farmers, if there are 3 to 4 days of consistent sunshine available. As such, the disparity in costs of mechanical and sun-drying erodes the business case for mechanical drying services because most farmers can only make money during the occasional harvest seasons when there is sufficient sunshine. In the last 2-3 years, many service providers sold their dryers, leaving NCPB as the primary provider of mechanical drying services.

Coping Mechanisms Adopted by Farmers

Given that sun-drying is not an option, and mechanical drying is uneconomical, farmers are adapting as follows:

- Delaying the harvesting of maize in anticipation that the rains will subside. However, this presents the risk of aflatoxin infestation, moulding and rotting on the farm;
- Harvest and immediately sell the wet maize to traders and miller who ensure that the price they offer the farmers is less the cost of drying.

5.3. Key Cost Drivers for Maize Milling

The survey found that maize grain was the most significant cost, accounting for approximately 68% of maize milling costs. Other cost components include; packaging costs (about 8% of milling costs) and handling costs which include cleaning, drying, storage and quality control (6.8% of milling costs).

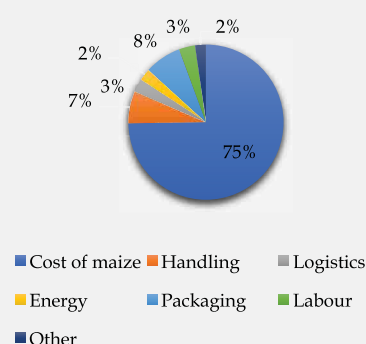
On average, millers incur an estimated KES 50 to produce a per kilogramme of maize flour, exclusive of other overhead charges (such as regulatory compliance, taxes, etc.). Table 8 below summarises the cost build-up for maize flour production.

Table 8: Cost build-up for maize flour production

Item	Per kg
Cost of maize grain	36.72
Handling	3.32
Logistics	1.31
Energy	1.26
Packaging	3.74
Labour	1.65
Other	1.11
TOTAL	49.47

Source: Survey findings, 2019

Figure 28: Distribution of milling costs



Source: Study findings, 2019



Considering that the cost of maize grain is the most significant cost component in maize flour production, reducing the cost of maize grain for millers bears the most fruit in terms of lowering milling costs and delivering more affordable flour to consumers. Thus, if the cost of maize grain delivered to millers would also reduce by approximately 50%, this would allow maize flour prices to retail at around KES 100 per 2kg packet, other factors remaining constant (see Table 9). This implies that significant efforts need to be directed, both by public and private sectors, to improve the productivity of Kenyan farmers and to reduce their transaction costs so that they can deliver cheaper maize to millers without compromising their incomes.

Table 9: Cost of milling maize flour, assuming a 50% reduction in maize costs

Item	Per kg (current scenario)	Per kg (hypothetical scenario maize price 50% lower)
Cost of maize	36.72	18.36
Handling	3.32	3.32
Logistics	1.31	1.31
Energy	1.26	1.26
Packaging	3.74	3.74
Labour	1.65	1.65
Other	1.11	1.11
TOTAL production costs	49.47	30.75
Millers Margin	10.00	10.00
Ex-factory price	59.47	40.75
Retail price	69.47	50.75

Source: Computations from field surveys, 2019

5.4. Key Policy-Related Challenges Affecting Maize Millers

This section analyses the business environment for maize millers in Kenya. It specifically focuses on regulatory compliance, compliance with quality standards, cross-border trade environment and government participation in domestic maize markets.

5.4.1. Regulatory Compliance

For this study, the cost of regulatory compliance covers compliance with tax and other administrative laws and regulations at both national and county level. This includes taxes and levies, as well as fees for various permits and licenses. The study found that maize millers are obliged to comply with a long list of regulatory requirements, all of which attract charges. These include, but not limited to the following:

Table 10: List of fees/permits/licenses required by millers

Value added tax (for some processors)	Business license	Distribution permits and fees
Agricultural produce cess	Parking fees for trucks	City Council license
Fee for Standardisation Mark by KEBS	Organisational Safety and Health Authority fees	Fortification fee/mark
Fire protection fee	NEMA certification	Food hygiene/handling fees
Health certification	Vehicle branding fees	

Source: Survey findings, 2019



Besides, the respondents spend between **KES 193,000 to 800,000** annually on permits and fees depending on the scale of their operations and distribution across the country. Some costs, such as cess and distribution fees are charged in all counties in which a miller either transits or distributes his products. For instance, if the distribution network of a miller covers ten counties, he/she will need to obtain 10 distribution permits per consignment. It is thus no surprise that all millers interviewed were unanimous that the overall cost of compliance with the various fees and licensing has increased significantly over the past five years, with some permits/fees rising by over 200% during that period.

The above list of regulatory requirements also highlights a large degree of duplication, which illustrates the extent for over-regulation of the milling sector. This calls for harmonisation of the regulatory requirements. For instance, fortification and quality/standardisation certification can easily be harmonised since fortification is also regulated by Standards which domicile at KEBS. Health certification and food hygiene/handling fees can also be harmonised, while the business permit can also absorb the vehicle branding fees.

Table 11: Millers' perception of most impactful regulatory requirements

Compliance area	Average rank of impact on business costs (1=smallest impact; 5=biggest impact)
Distribution permits/fees	4.88
Business License	3.25
County cess	2.4
KEBS certification	2.25

Source: Survey findings, 2019

From the above list of charges, levies and permits, the survey further found that distribution permits were perceived by millers to have the highest impact on their cost of doing business, followed by business licensing costs, county cess and KEBS certification (**Table 11**).

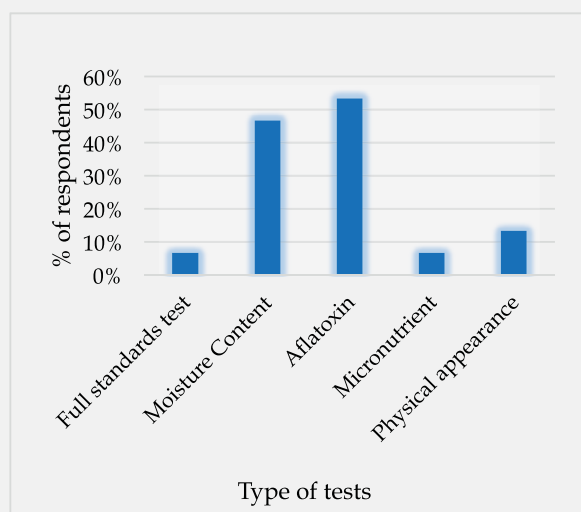
5.4.2. Compliance with Quality Standards

Over 75% of respondents exhibited awareness of Kenya (East African) Standards for grains and flours and claimed to use them in their day-to-day business operations. Furthermore, some millers also comply with international standards for quality management (ISO 9001), environmental management (ISO 14001) and food safety certification (FSSC 22000). 80% of the millers also claim to use laboratory testing for quality control of grains and flour, while 33% also use observation methods to determine quality, particularly during the purchase of grains. The survey also found that approximately 40% of respondents owned an in-house laboratory, while 60% of outsourced testing services to third parties. The conventional laboratory service providers from the survey are Jomo Kenyatta University of Agriculture and Technology, University of Nairobi and the Interfield Testing Laboratory.

Interestingly, few millers run full standards tests (grading) when purchasing grains. Instead, most of them conduct aflatoxin and moisture content checks only (Figure 29). For millers using either in-house or outsourced laboratory services, micronutrient tests are equally rare, while physical observation is also used to complement the lab tests. The tests on maize flour are also generally limited to aflatoxin, moisture content and vitamins/micronutrients (Figure 30). However, considering that maize fortification is mandatory in Kenya, the level of testing of vitamins and nutrients was expected to be higher. Furthermore, the respondent rate for testing for any of these parameters should ideally be 100% to guarantee consumers of safe and quality food. The fact that less than 60% of respondents test for aflatoxin means that a significant proportion (over 40%) do not conduct tests, which presents a significant food safety risk for consumers.

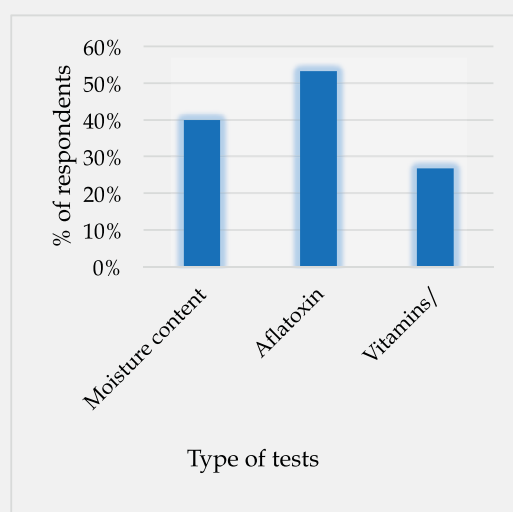


Figure 29: Common quality control tests conducted by millers during the purchase of maize grain



Source: Field Survey, 2019

Figure 30: Common quality control tests on maize flour



Source: Field Survey, 2019

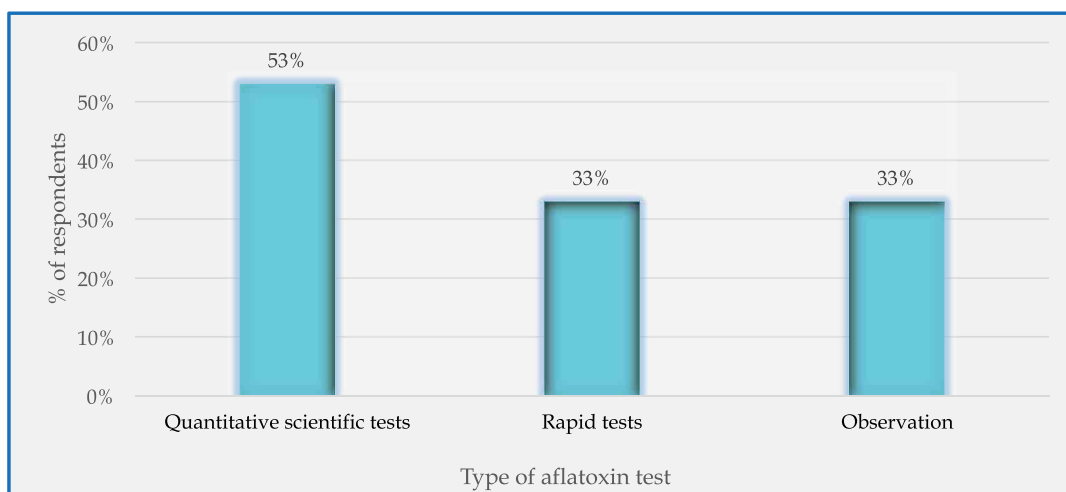
Despite most millers owning or having access to food laboratories, the low prevalence of full standards checks, particularly during the purchase of maize grain, means that there is limited grading of commodities in the value chain. It further suggests that decisions to buy maize are not based on the grade of a particular consignment, but exclusively on the degree to which the shipment meets the minimum food safety criteria which are standard across all grades in the Kenya (East African) Standards for maize. The market does not offer prices based on quality implying that as long as the commodity is sufficiently dry and is of acceptable aflatoxin levels, then the price offered is the same irrespective of the other quality parameters such as discolouration, immature and shrivelled grains, broken kernels, foreign matter, etc. If more millers were to offer different prices for different quality of grain as per the standards, this could create a trickle-down effect to farmers to incentivise them to invest in improved post-harvest handling and storage of maize to earn higher incomes for superior quality grains.

Aflatoxin Control

Maize flour has been implicated in aflatoxin poisoning resulting from mould contamination of maize on the farm or in storage. In Kenya, most human exposure to the toxin is through consumption of maize due to heavy reliance on this grain as a dietary staple (Abizari et al., 2016). Ensuring aflatoxin levels in flour below 20 ppb assures the safety of the product to consumers (Flanders et al., 2011). The frequency of internal monitoring of maize flour among the mills varied greatly among the mills. Majority of millers (53%) use quantitative tests and equipment such as ELISA® and High-Performance Liquid Chromatography (HPLC) machines. However, a significant proportion also uses qualitative rapid test kits and observation methods (33% each) as highlighted by **Figure 31**.



Figure 31: Test methods for aflatoxin used by millers



Source: Field surveys, 2019

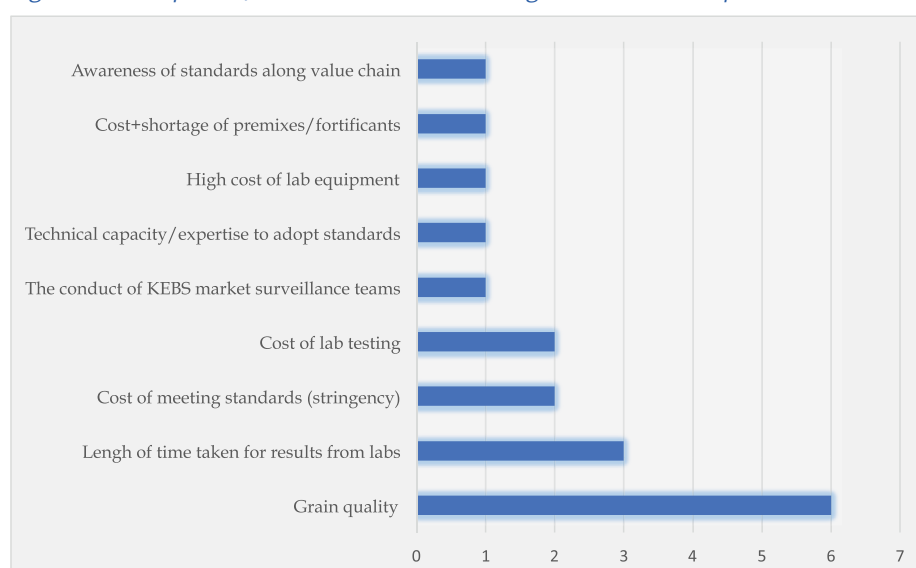
Since aflatoxin contamination tends to be concentrated in certain geographical areas, and because levels can become very high under traditional storage conditions, smallholder producers in high-risk areas are at greatest risk of high levels of aflatoxin exposure. Deaths and acute sickness due to aflatoxin poisoning in Kenya have been linked to maize grown and stored by smallholder farmers (Azziz-Baumgartner et al., 2005). Because of the possible negative effects of aflatoxin exposure on child development on one hand, and broad population-wide impact through maize on the other, control of aflatoxin in maize should be prioritized.

Hermetic storage bags, designed to reduce insect pest losses, have been shown to reduce aflatoxin contamination by between 40% and 66% after three months of storage, depending on the starting moisture content, level of contamination, and other environmental factors (Walker et al., 2018).

Challenges in Quality Control

The biggest challenge facing millers regarding enforcing standards is the quality of the grain. Which, in most cases, was affected by moisture content and aflatoxin contamination, especially during harvest seasons that coincided with heavy rainfall (**Figure 32**). Other challenges highlighted included the length of time taken to receive test results from laboratories, the cost of the tests and the cost of compliance.

Figure 32: Perception of millers on common challenges in standards implementation



Key: 1= Not a challenge 7= significant challenge

Source: Field survey, 2019



5.4.3. The Cross-Border Trade Environment

The study also assessed perceptions of millers on the cross-border trade environment considering the importance of grain imports from neighbouring countries to supplement local supplies as elaborated earlier in this report. In this area, two main aspects of cross-border trade were assessed - the time taken and the cost incurred to move maize grain across borders.

The study found that approximately 40% of respondents perceived that the time taken in moving maize grain across borders has improved over the past five years. However, a similar proportion pointed out that the cost of doing business across borders has increased during the same period. This shows that investments to facilitate trade, such as One-Stop Border Posts have succeeded to reduce the time taken to clear goods across borders but have not reduced the cost of conducting business.

5.4.4. Government Participation in Domestic Markets

Millers generally do not appreciate the participation of government in maize markets because in most cases, the policy instrument the government uses is setting the floor price for maize grain, which results in negative implication on milling costs. During periods of adversity, the government has intervened by selling maize reserve stock to millers at subsidized prices. Forty percent of respondents reported that government intervention in markets has made business worse over the past five years (that is, the modality of government participation in markets has deteriorated during this period), with only 20% acknowledging that the modalities for government intervention have improved favourably from the perspective of millers.

Aspect of government involvement in markets	The perception amongst millers (1=highly unfavourable, 5=very favourable)
Price determination for government purchase of maize from farmers for food reserves	2.1
Allocation of food reserve stocks to millers	2.2
Price at which food reserve stocks are sold to millers	3.4
Quality of maize sold from food reserve stocks to millers	1.9
The overall perception of government involvement in markets	3

Source: Field Surveys, 2019

Implication of the Business Environment on Milling Costs

Millers were unanimous that the costs of milling have increased over the past five years. The most important factors behind this increase are (in order of importance) compliance with regulations (fees, taxes and permits), government participation in markets through the setting of the floor price for maize grain, compliance with quality standards and cross-border trading costs. Other areas which affect milling costs but to a lesser extent are costs of energy and trade finance.

Table 12: Ranking of key business environment factors for maize milling

Issue	Rank (1=most important)
Compliance with regulations, fees, levies and taxes	1
Government participation in markets	2
Compliance with Standards	3
Cross-border trade costs	4

Source: EAGC, 2019



6. Conclusions and Recommendations





6 Conclusions and Recommendations

6.1. The Key Findings from the Study

Demand and Supply

- Maize plays a central role in contributing to food security, trade and livelihoods of millions of people in Kenya. Maize is the cheapest source of calories among the Kenyan households accounting for 24 per cent of all calories consumed and 50 per cent of staple food calories consumed.
- The demand for maize currently stands at 52 million bags, and it has been projected to reach 60 million bags by 2025. Kenya is a net importer of maize due to higher consumption relative to domestic production. The average maize production over the past 15 years has been estimated at 3.2 million metric tons while consumption was 3.5 million metric tons.

Gross Margin Analysis for Maize Production

- Large-scale farmers are 33% more productive than small-holder farmers, with large-scale farmers achieving yields of 24 bags per acre compared to 18 bags per acre by smallholder farmers¹². This means that large scale farmers earn almost 39% higher profits than their smallholder counterparts despite incurring 31% higher production costs.
- The three biggest cost drivers for maize production irrespective of the production system are fertilisers, machinery and labour. Collectively, these costs account for 64% of total production costs in large-scale systems and 77% in smallholder production systems. However, labour is a far bigger cost driver for smallholder farmers (36% of total production costs) compared to large-scale farmers, most likely due to limited mechanisation in smallholder farming systems.

¹² Tegemeo (2018) define a large-scale farmer as one who farms 50 or more acres of maize, while a smallholder farmer cultivates less than 10 acres of maize. Both types of farmers have 1 season of maize



Maize Trade

- Cross-border trade flows are aligned to the respective crop calendars for Kenya and its main trading partners (Uganda and Tanzania). Peak imports are usually during the second and third quarters of the year, which coincide with the lean period domestically and harvests from trading partners. Imports from Uganda typically peak in the third quarter while those from Tanzania peak during the second quarter.
- Informal cross-border trade usual surpassing the volumes that are sold through the formal channel. For instance, in 2017, Kenya informally imported 368,167MT of maize from Uganda. Kenya like the other EAC states has adopted the harmonized EAC standards for maize, however, implementation and enforcement is limited.

Post-Harvest Losses

- Post-harvest losses in maize are estimated at 8% - 40% according to different reports. Regarding the value chain structure as a percentage of total production, most losses happen at storage (9%) due to pest infestation, moulding caused by inadequate drying, rodents and poor hygiene. Smaller amounts are also lost during shelling (1.8%), drying (3%) and harvesting (2.2%).
- Maize flour has been implicated in aflatoxin poisoning resulting from mould contamination of maize on the farm or in storage. Hermetic storage bags, designed to reduce insect pest losses, have been shown to reduce aflatoxin contamination by between 40% and 66% after three months of storage, depending on the starting moisture content, level of contamination, and other environmental factors.

Maize Milling

- Maize milling is dominated by small and medium scale millers, who are estimated to control approximately 60% of the market, with an estimated milling capacity of not more than 150MT per day. Large-scale millers account for approximately 40 % of the market with an installed milling capacity of approximately 6,600MT per day.
- Hammer mills (posho millers) have a milling capacity of 6 -110 metric ton per month, depending on the location of the mill. Also, some posho millers had invested in maize dehullers to produce refined flour commonly referred to as “Number 1”. These mills are mainly found in rural areas and urban slum areas. The main challenge facing these millers is limited capital for expansion. Additionally, most of them are use diesel as the source of energy. Due to the volumes that are milled, there is underutilization of the fixed costs, such as machines and labour.
- Small-scale millers are estimated to be about 50–75 in number with an installed capacity ranging from 10 - 200 metric tons per month. These millers produce sifted maize meals and maize by-products such as the composite maize meal that they provide in their own consumer brand packaging. The major constraints faced by the small-scale millers is the use of poor technologies which affects the quality of their products. The other constraints are underutilized processing capacities, limited access to finance, the unreliability of grain supply and high cost of power.
- Medium and large-scale millers account for about 93% of the annual total installed milling capacity, estimated at 1.4 million metric tons. Most of the mills are located in Nairobi and Mombasa. Both cities account for 60% of the national installed capacity, followed by Eldoret with 15%, then Thika, Kisumu and Kitale with the remaining installed capacity. The four main large-scale millers account for 70% of total milling capacity. Apart from sifted maize meal, these millers are also involved in animal feed production.
- Cost of milling maize grain: The cost of maize grain accounts for approximately 75% of milling costs, packaging costs (about 8% of milling costs) and handling costs which include cleaning, drying, storage and quality control (7% of milling costs).



- High cost of post-harvest handling and storage: For instance, for farmers in Nakuru, mechanical drying accounts for 56% of the post-harvest handling costs as opposed to sun-drying, which accounts for 34%. Sun-drying is only practical for small quantities of grain. This implies that a farmer will incur a cost of over KES 500 per bag on storage and handling if they harvest maize with a moisture content of 20%.
- Low enforcement of quality standards for maize grain: Maize grain in the domestic market is not priced based on quality consideration. In most cases, the grain is affected by moisture content and aflatoxin contamination, especially during harvest seasons that coincided with heavy rainfall. Other challenges to limiting enforcement of standards include the length of time taken to receive test results from laboratories, the cost of the tests and the cost of compliance.
- Cost of doing business for millers has increased over the past five years, driven mostly by substantial increases in regulatory compliance costs (taxes, fees and permits), costs of compliance with standards (sourcing quality grain and meeting standards requirements for milled products), government participation in markets through food reserves and increase in costs of cross-border trade despite the reduction in time taken to clear goods across borders.

6.2. Recommendations for Reducing Maize Milling Costs and Improving Food Safety

The study recommends the following:

6.2.1. Policy Recommendations

a) **Proposals to reduce the cost of production of maize**

Considering that the cost of maize accounts for approximately 75% of the total cost of production of milled maize products (maize flour), reducing the cost of production of maize is critical for reducing the cost of maize flour milling and food prices for consumers. The study thus puts forward the following recommendations to this effect:

- i. **Increase maize productivity:** By reducing the yield gap, through the following interventions:
 - a. **Improve extension services to smallholder farmers:** Under the devolved system of government, extension services are under county governments. Therefore, counties need to allocate more financial resources to extension services to recruit, equip and facilitate extension workers. Extension services will help farmers to improve their agronomical practices, including but not limited to the application of fertilisers and agro-chemicals, climate-change adaptation and soil health management.
 - b. **Improving and expanding the fertiliser subsidy programme:** A fertiliser subsidy programme has been in place since 2008 but has achieved mixed results. Amongst its challenges are inadequate targeting of beneficiaries whereby the programme does not reach some smallholder farmers and delayed deliveries of fertilisers to farmers due to procurement and distribution challenges, among others (EAGC, 2018). The government subsidy programme can be improved by scaling up the E-Voucher System that is being implemented in the KCEP-CRAL project. Under the E-Voucher System, both farmers and agro-dealers should be recruited and vetted prior to being onboarded on to the subsidy programme. The agricultural inputs include fertiliser, seeds, agro-chemicals, agro-chemicals, hermetic bags and tarpaulins. Farmers can then use the e-vouchers to obtain inputs at subsidised rates through the vetted agro-dealers, who then redeem the vouchers through participating banks (Equity Bank and Co-operative Bank). This allows farmers to access all the necessary inputs for increased farm productivity, and basic post-harvest handling and storage thus reducing on-farm post-harvest losses.
 - c. **Additionally, the provision of inputs should be linked with the agro-ecological conditions,** including soil health and not have a “one-size-fits-all” approach. For instance, where soils are acidic, then the agro-input package should ideally include lime and allow a farmer to access the right fertiliser given the nutrient profile of his/her soil.



- d. **Support bulk access to farm inputs and farm mechanisation services:** Smallholder farmers are high-cost producers due to the absence of economies of scale. Nonetheless, these economies can be attained by the collective purchase of inputs and mechanisation services by smallholder farmers. For instance, EAGC has supported farmer groups in Uasin Gishu and Trans-Nzoia to buy fertiliser directly from Toyota Tsusho East Africa Limited (Baraka Fertiliser) at KES 2,800 per 50kg bag, with free transport to their farms, against a market price of KES 3,200 to 3,500 per 50kg bag. Through a similar process, other groups were able to buy hermetic storage bags directly from manufacturers at KES 190 per unit compared to retail prices of KES 250 per unit.
- ii. **Curtail land sub-division in maize production areas:** Land sub-division makes maize cultivation increasingly unprofitable for farmers because economies of scale cannot be attained through small parcels of land. Therefore, appropriate policy reforms should be initiated to ring-fence agricultural land and thus protecting it from sub-division and from uses that are outside agriculture (such as real estate).

b) Proposals to improve the market system for maize trade through structured trading systems

Maize trading is largely informal and unstructured at a national level whereas the market is not properly coordinated at the regional level. Furthermore, post-harvest losses due to low adoption of handling and storage technology along the value chain increase transaction costs, which are reflected in the cost of maize grain.

To improve efficiency, there is need for wide-scale adoption of structured trading systems for grain trading. The Warehouse Receipt System (WRS), whose legislation was enacted in June 2019, forms the basis for structured grain trade. The WRS allows storage of commodities in a certified warehouse – that is, a facility with the necessary facilities and management systems to guarantee the quality and quantity of commodities stored therein. A Warehouse Receipt confers ownership of goods to the depositor (farmer). It also becomes a valid document of title which can then be used as collateral to obtain financial services for trading purposes whereby buying and selling of Warehouse Receipts transfers ownership of the underlying commodities without physically relocating the goods. The WRS underpins commodity exchanges and trade platforms which ultimately facilitate trade in commodities in an open and transparent manner. In Kenya, the Kenya Multi commodity Exchange (KOMEX), the NeXT Commodity Derivatives Exchange of the Nairobi Securities Exchange and the EAGC G-SOKO Grain Trading Systems are amongst the systems available for structured trade in maize and other agricultural commodities.

The WRS, trade platforms and exchanges thus offer the following benefits to millers:

- **Cost reduction for millers:** Millers can purchase maize by directly buying Warehouse Receipts through the exchanges and trading platforms and take physical delivery at a convenient time depending on their operations. This reduces the need for stockholding at a miller's premises, thus reducing storage costs and associated overheads.
- **Quality and quantity guarantee:** The exchanges, together with the underlying WRS principles, guarantee the quality and quantity of commodities stored and transacted therein. This helps to reduce the need for stringent test protocols by millers and thus reduce their costs.
- **Reduce multiple handling of grains between farmer and miller:** Bulk aggregation and transacting on the exchanges means that the miller's supply chain is shortened as they
- **Price risk management and procurement planning:** Millers can purchase forward and futures contracts on commodity exchanges for delivery of goods in the future. These trading instruments are useful for hedging against price risks and allow millers to plan ahead in terms of purchases of maize against milling requirements based on other operational and market factors.



c) Tax proposals to reduce maize milling costs

Given the relatively high cost of maize in Kenya, it is imperative to find creative ways to reduce the cost of maize beyond improving productivity and reducing on-farm costs. The study thus proposes two tax-related measures as follows:

- i. **Remove taxes on post-harvest handling equipment, products and services:** Post-harvest handling equipment, products and services attract Value Added Tax (VAT). In addition, some imported post-harvest handling products, such as hermetic storage technologies attract import duty unless exemption letters are obtained by importers. This is despite the fact that post-harvest losses in maize value chain estimated at approximately 30% annually. This equates to approximately 13.4 million 90kg bags out of the 44.6 million bags produced in 2018¹³, and between 15% and 19% of post-harvest losses being caused by pest infestation (FAO, 2014). Therefore, these taxes undermine the uptake of good post-harvest handling practices, particularly by farmers.
- ii. **Abolish multiple cess and similar charges of maize consignments:** Agricultural produce cess and similar levies are charged by county governments on agricultural produce through County Finance Acts. Cess on agricultural produce presents a marketing and trade challenge because its rate varies across counties and consignments attract multiple charges during transportation from origin to destination counties (CGA, 2017). For instance, a consignment of maize from Trans-Nzoia to Nairobi will attract cess in at least 6 counties (Trans-Nzoia, Uasin-Gishu, Kericho, Nakuru, Kiambu and Nairobi). Assuming a constant cess fee of KES 50 per bag in each county, this amounts to KES 300 per bag paid as cess. Produce cess thus represents a significant cost for maize trade and milling.

It is proposed that the rate of cess is constant across counties and applicable only one at the source of origin. Furthermore, cess should nonetheless be tied to a service provided to value chain players by county governments. Similar to (b) above, grain industry stakeholders thus need to engage in concerted and consistent efforts to reform cess collection by county governments.

6.2.2. Institutional Recommendations

a) Public Private Partnerships to reduce post-harvest losses and improve grain handling and marketing in the maize value chain

Post-harvest losses in the maize value chain are high, mostly due to low uptake of handling and storage technologies. At the same time, public bodies such as NCPB have large, underutilised storage capacity which could help alleviate these challenges. This presents an opportunity for public-private partnerships to alleviate post-harvest challenges. The study thus recommends four public-private partnership approaches that can be considered to improve post-harvest handling, storage and marketing of maize.

13 Provisional production estimates provided by the Economic Survey 2018 by KNBS.



- i. **Restructuring NCPB into a public-private commercial entity:** In this approach, NCPB can be restructured into a Public-Private Partnership limited company whereby the private sector investors own the majority of the shares while the government holds minority shares, similar to the Kenya Multicommodity Exchange ownership structure. This will allow the private sector to inject the much-needed capital to modernise storage facilities that are currently not in optimal shape for grain storage. This will also allow the private sector to run the organisation as a pure commercial entity with minimal political interference, enabling it to provide more efficient storage and handling services for both the government and private sector players. This may be attractive to millers who are currently obliged to invest in their own storage infrastructure despite grain storage not being their core business. With a well-functioning, commercially-run NCPB, millers can outsource maize storage to the privatised NCPB, thus creating a win-win situation whereby millers reduce significant overheads associated with in-house grain handling and storage while the new privatised NCPB earns revenue streams from storage and handling fees. This is similar to the model deployed in wheat where wheat millers import bulk consignments of wheat, store the same at customs-bonded warehouses operated by Grain Bulk Handlers Limited and only transfer wheat to their mills based on milling requirements.
- ii. **Engage the private sector for storage of food stocks on behalf of Strategic Food Reserve (SFR):** Engaging private sector participation in managing the food reserves stocks. For instance, the use of bulk hermetic storage of grains which are much more efficient and cost-effective compared to traditional warehouse or silo storage. Bulk hermetic storage is already extensively used in South Africa, Israel and India, and can easily be adopted in Kenya.
- iii. **Leasing NCPB facilities to private sector operators:** There is already interest among the private sector to lease NCPB facilities for either own storage or to provide third-party storage services. In this case, the private sector lessee can enter into medium or long-term lease contracts with NCPB and invest to improve the leased facilities from which they will offer storage and handling services. This approach complements the proposed operating model by the SFR which allows it to enter into agency agreements with private sector players for storage of grain for food reserve purposes. It also allows NCPB to commercialise and modernise its idle capacity and thus contribute to its overall sustainability.

b) Data and Information to inform policy and business decisions

Data and information are critical for timely and accurate policy and business decision-making. Despite its importance, data collection is costly, data contributors raise concerns over confidentiality and access to such data is limited.

The Ministry of Agriculture with support from development partners is taking measures to improve this situation by revamping the Kenya Food Balance Sheet. This is by convening public sector, private sector and humanitarian organisations to share their data on commodity stocks and trade movements monthly, with maize being the priority commodity.

Nonetheless, the government could consider adopting the South African model of data collection whereby the South African Grain Information Service (SAGIS), an independent, non-profit, private sector organisation, is empowered by law to collect and synthesise market information on grains. In this model, the so-called “**Co-Workers**”, which are essentially grain value chain players (farmers, domestic and international traders, silo operators, millers, wholesalers, etc.) are required by law to submit their stocks data to SAGIS weekly for consolidation. On the other hand, SAGIS maintains complete autonomy from the government and is thus able to assure total confidentiality and integrity of the data collection system. The consolidated, anonymised data from SAGIS is then availed to both public and private sector players to inform their respective policy and business decisions. This process has created transparency and trust between the public and private sector in the management of food security.



6.2.3. Legal and Regulatory Recommendations

a) Regulatory and compliance improvements for food safety in the maize value chain

Grain sector stakeholders pointed out that the regulatory burden for ensuring food quality and safety to consumers is predominantly and perhaps unfairly placed on millers only with little regard being paid to upstream value chain players such as producers and traders. To address this, three regulatory and compliance models can be considered:

- i. **Establishing regulations for mandatory food testing by all food handlers**, whereby food handlers, described as value chain players that transact in food commodities such as aggregators, warehouse operators and traders, will be required to conduct quality and safety tests on commodities purchased and sold, and keep auditable records of the same. Regulations could cover the following areas:
 - Compel food handlers to either invest in in-house quality assurance (i.e. a laboratory plus a trained and certified grader) or outsource such services to competent service provider.

Provision for training, examination and certification of graders, such that only certified graders will be allowed to sample, test and grade commodities. This process will essentially make commodity grading a regulated profession due to its sensitivity and importance in assuring food safety.

This is expected to reduce the regulatory pressure on millers and oblige other value chain players to improve quality control of grains. It will also help to stimulate the determination of market prices according to grades of traded commodities.

- ii. **Establishing a private sector self-regulation mechanism to enforce the standards:** This is where grain industry bodies such as EAGC develop a regulatory mechanism where its members and other value chain players will monitor their own compliance to quality and food safety standards, rather than have government agencies monitor and enforce the standards.
- iii. **Improving compliance support:** To achieve food safety, there is a need for all stakeholders, from farm to fork, to work together towards delivering safe food to consumers. However, this is largely hampered by disjointed activities by actors within the value chain. It is proposed that the regulators provide capacity building, diagnostics and any other support required to deliver safe food to the consumers, as opposed to their current policing approach.

b) Regulatory improvements to reduce multiplicity of regulatory functions in the maize value chain

The study found multiplicity of regulatory functions as the biggest hurdle to improving the business environment in the maize value chain. The maize milling industry is currently regulated by multiple regulators including KEBS, Public Health under the national government and public health offices within the 47 counties. Multiple regulation negatively affects the ease of doing business in addition to increasing the cost of compliance. Moreover, it creates unpredictability in the business environment.

The study thus recommends **rationalising of regulatory requirements** whereby food safety and quality matters will be regulated nationally from the Public Health and Kenya Bureau of Standards under the national government without requiring additional licensing or permits for the same matter at county level.





7 Implementation Matrix

	Policy	Action	Responsible Institution	Short Term	Medium Term	Long Term
1	Improve the market system for maize trade through structured trading systems	Adoption of structured trading systems for grain trading	Government of Kenya- State Department of Trade Nairobi Stock Exchange- Kenya Multi Commodity Exchange (KOMEX)		X	
2	Tax proposals to reduce maize milling costs	Allowing duty-free yellow maize specifically for animal feeds under duty remission scheme	Government of Kenya - KRA, MoALF&C	X		
		Removing taxes on post-harvest handling equipment, products and services	Government of Kenya - KRA, MoALF&C	X		
		Abolishing multiple cesses and similar charges of maize consignments	Government of Kenya and County Governments		X	
3	Public-Private Partnerships to reduce post-harvest losses and improve grain handling and marketing in the maize value	Restructuring NCPB into a public-private commercial entity	Government of Kenya		X	



	Policy	Action	Responsible Institution	Short Term	Medium Term	Long Term
		Engage the private sector for storage of food stocks on behalf of SFR	Government of Kenya		X	
		Leasing NCPB facilities to private sector operators	Government of Kenya		X	
		Purchase of maize for the SFR through trade platforms and commodity exchanges	Government of Kenya, Industry Players		X	
4	Strengthen data collection and information sharing to inform policy and business decisions	Improve coordination of data collection, and data sharing	Government of Kenya- KNBS, County Governments, Research Organisation, Universities, EACG, Industry Players		X	X
5	Strengthen implementation and compliance mechanisms of quality standard for improved food safety along maize value chain	Strengthen the provision of extension services at National and County levels Promote self-regulation to enforce adherence to food safety standards	Government of Kenya- KNBS, County Governments, Research Organisations, Universities, EACG, Industry Players	X	X	

This is an extract of the report on The Maize Value Chain in Kenya: Implications on Maize Milling Costs





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Annexes

Annex 1: Summary of Respondents

Respondent category	Location	Number of respondents	Type of info/data collected
Maize farmers (farmer groups)	Nakuru	5 respondents	· Production costs and key cost drivers · Post-harvest handling and storage practices and costs · Marketing of maize
Maize farmers associations	Nairobi	1 respondent	· Business environment for farmers · Production costs and key cost drivers · Post-harvest handling and storage practices · Compliance with grain standards · Marketing of maize
Maize traders	Nairobi	5 respondents	· Business environment for maize trade · Logistics costs and considerations · Post-harvest handling and storage practices · Compliance with grain standards · Trade financing perspectives · Maize marketing and trade
Maize processors/ millers	Nairobi, Mombasa, Thika, Ruiru, Bungoma, Eldoret and Meru	SME Millers: 15 respondents Large millers: 2 respondents Total: 16 respondent companies	· Business environment for maize milling · Compliance with grain standards · Maize marketing and trade · Milling costs and key cost drivers
Millers and Manufacturers' associations	Nairobi and Muranga	2 respondents	· Business environment for maize milling · Compliance with grain standards · Maize marketing and trade · Milling costs and key cost drivers
Post-harvest solution providers	Nairobi	3 respondents	· Available post-harvest solutions · Cost drivers for post-harvest solutions · Opportunities and challenges presented by post-harvest solutions







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